Online Appendix
International Trade Liberalization and Domestic Institutional Reform:
Effects of WTO Accession on Chinese Internal Migration Policy

Yuan Tian*

January 2022

*School of Economics, University of Nottingham (yuan.tian1@nottingham.ac.uk)

Online Appendices

(Not for publication)

A Additional Information on Migration Regulation Data

A.1 The Number of Regulations in Different Stages of Study

Table A1: Summary of the number of regulations

Number of migrant-related regulations	**Total number**	**Prefecture level**	**Province level**
Full sample from keyword search, 1978 to 2016	4,273	2,915	1,358
Hand coding			
by the author	4,273	2,915	1,358
by two research assistants	686	390	296
NLP coding			
used as the training sample in NLP	686	390	296
coded using NLP	4,273	2,915	1,358
Out of the full sample, all 340 prefectures			
1978 to 1995		69	
1996 to 2001		144	
2002 to 2007		748	

Note: This table presents the number of regulations used in different stages of study.

A.2 Three Supervised Learning NLP Methods and Results

A.2.1 Introduction of the Three Classification Methods

The logic of the supervised learning NLP method is to make use of the entire text of the regulations, and build a model based on the elements of the text, train the model with some regulations with scores coded, and then code the rest of the regulations using the model.

There are three NLP methods I consider. First is a decision tree. Here, only the words and word frequencies are used. The advantage is that it is easy to summarize the decision rule. i.e., what words are important in making the decisions. The disadvantage is that only words and word frequencies are used.

The second is multilayer perceptron classification with neural networks (MLP). The advantage is that the decision rule is more complicated than the decision tree. The disadvantage is that only words and word frequencies are used, and that it is less transparent in what words are important and how decisions are made.

The last one is classification based on word embedding. The advantage is that the entire text can be used, including the neighbors of the words. Each word is turned into a vector based on its context, and similar words will be represented as similar vectors. Then the vectors are used to classify the regulations. The disadvantage is that usually a large training dataset is needed, and the decision rules are less clear.

A.2.2 Method 1: Random Forest

The choice of parameters in the random forest algorithm is as follows: 200 trees, 101 attributes to draw from, and gini split rule. Results are not very sensitive to the choice of the parameter values. To avoid overfitting of a simple decision tree based on a training set, the random forest introduces randomness in the sampling method for the training set and the attributes used. First, the random forest uses a bootstrap sample of the same size of the training set using sampling with replacement. This means that the training set is different for each tree. Second, the random forest draws a sample of the attributes to choose from in each step of growing the tree. Due to the randomness of the training set, the model accuracy can be evaluated as the share of mis-classified regulations that are not included in the training process, which is called the out-of-bag (OOB) prediction error. The average OOB prediction error across the 200 trees is 20.70%. The classification outcome is determined by a majority vote across the 200 trees.

Among all the regulations, 3,092 regulations generate the same score as my own coding, 538 regulations have 1 unit larger coding, 569 with 1 unit smaller, and 74 regulations differ from my coding by more than 1 unit. The cross-tabulation results are in Table A2. One pattern is that hand coding seems to be more nuanced, with more –2, –1, and 2, while the random forest coding has more 1. This could reflect the possibility that since the majority of the regulations have either 0 or 1 according to hand coding, the random forest model has more training samples in those categories to correctly specify their characteristics. For the ones with a smaller sample size, the performance of the random forest model can be compromised. Overall, the two sets of codings are highly correlated, with a correlation of 0.73.

Table A2: Cross tabulation of random forest coding and manual coding

	Random forest coding					
Manual coding	–2	–1	0	1	2	Total
–2	2	1	4	4	0	11
–1	0	6	19	8	0	33
0	0	12	1,112	351	10	1,485
1	0	0	181	1,180	167	1,528
2	0	0	48	376	792	1,216
Total	2	19	1,364	1,919	969	4,273

Note: This table presents the comparison between the random forest coding and the manual coding, by showing the number of regulations in each bin.

Another way to evaluate the coding by random forest and by hand is to see if these codings differ systematically by topic categories. Figure A1 shows the distribution of the gap between the hand coding and random forest coding results by topic. Overall, hand-coding seems to code the "social insurance" and "work" type more positively. On the other hand, random forest coding favors the "other" category.

Figure A1: Difference between hand coding and random forest coding results, by topic

Note: This figure presents the comparison between the random forest coding and the manual coding, by showing distribution of the gap between the hand coding and random forest coding results by topic.

Figure A2 presents the most important words used in the random forest classification.

Figure A2: The most important 200 words used in the random forest algorithm

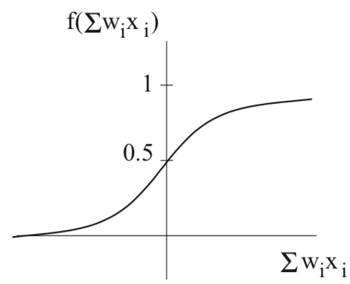
Note: This figure shows the most important 200 words used in the random forest classification. A larger font indicates a higher level of importance. For additional details of the method, see Appendix A.2.2.

A.2.3 Method 2: Multi-Layer Perceptron Neural Network Classification

The second algorithm is multi-layer perceptron (MLP) neural network. First, regulation texts are split into words. The words with highest frequencies (e.g., the top 1000 words) will be chosen to form the vocabulary. Then each regulation will be represented as a matrix, with the word vector in the vocabulary combined with an indicator variable equal to one if the word shows up in the regulation. Second, I build a MLP neural network, with the regulation text matrix as the input signals, the score of migrant friendliness as the output signals, and one hidden layers with neurons that are fully-connected with the input and the output layer, or more hidden layers. Transfer functions take weights on signals and neurons to calculate the value of neurons on the next layer. Figure A3 presents a neural network with two layers, the first layer takes the input signals and connects them to the hidden neurons, and the second layer takes the hidden neurons to output neurons. Eventually, output neurons projects out the output signals. For example, Figure A3 shows a popular transfer function (sigmoid) that takes the weighted sum of x_i and projects it to a value between zero and one. Weights are randomly assigned at the beginning, and during the training process, updated based on the feedback of prediction results. The model is trained to have low loss and high prediction accuracy.

Figure A3: Depiction of a popular transfer function and an example neural network with two interconnected layers

Fig. 5.1 A popular transfer function: the `sigmoid`



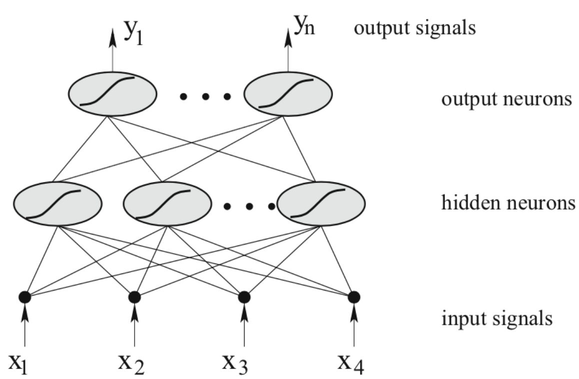


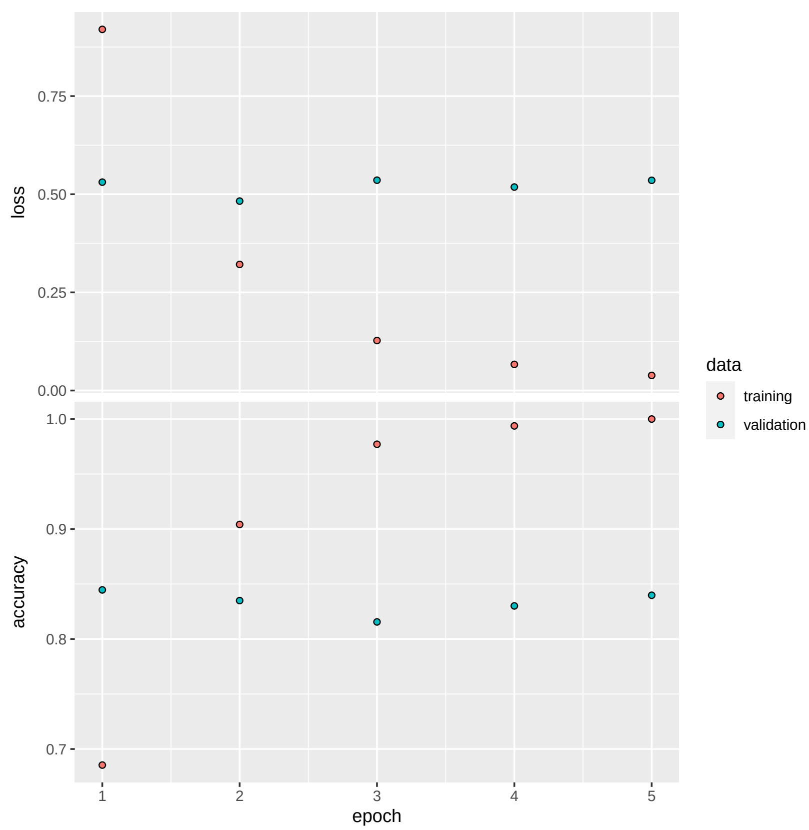
Fig. 5.2 An example neural network consisting of two interconnected layers

Note: This figure is from Kubat [2017]. The sub-figure on the top right corner depict a popular transfer function (sigmoid) that takes the weighted sum of x_i and projects it to a value between zero and one. The bottom sub-figure depicts an example neural network consisting of two interconnected layers, where the input signals are transformed to hidden neurons and then to output neurons.

Again, I use the 686 regulations that are coded by three independent people as the training set. The most frequent 2000 words in these text are chosen as the vocabulary. Out of the 686 regulations, 70% are used for actual training and 30% are used for model validation. One epoch is one round of model training using the regulations in the training set, and five epochs are used. Other parameters include: 512 neurons are used in the hidden layer, 20% of the neurons are dropped before feeding into the transfer function to avoid overfitting, 32 regulations are used to train the model before the weights updates, and categorical crossentropy is used as the loss function.

The results of the training process is shown in Figure A4. For example, in Epoch 1, the training accuracy is below 0.7, indicating that when the model is applied to the training regulations, less than 70% have the correct coding. The validation accuracy is about 0.85, indicating that when the model is applied to the validation regulations, about 85% have the correct coding. Comparing Epoch 1 results with those of later epochs, we see that the training accuracy improves and arrives at 100% in Epoch 5, and the validation accuracy remains stable. This is because the algorithms adjust over the epochs to improve training accuracy, but it does not necessarily help the validation accuracy since the validation regulations are not used in the training process.

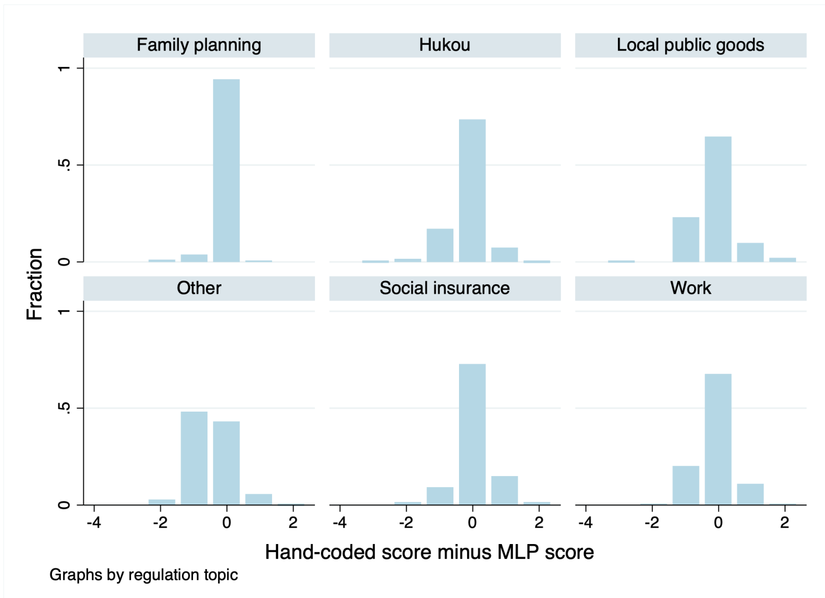
Figure A4: MLP model results



Note: This figure shows the training process of the MLP model. The horizontal axis depicts the epoch, where one epoch is one round of model training using the regulations in the training set. The red dots represent the training data, and the green dots represent the validation data. The vertical axis in the top figure shows the value of the loss function, and the vertical axis in the bottom figure shows the value of accuracy. In this analysis, categorical crossentropy is used as the loss function. The smaller the loss, the better the model fit. Accuracy measures the opposite: the higher the accuracy, the better the model fit.

The estimated MLP model generate predicted scores very close to my own codings when I use it to predict the migrant friendliness of the 4,273 regulations. Among all the regulations, 2,959 regulations generate the same score as my own coding, 873 regulations have 1 unit larger coding, 362 with 1 unit smaller, and 79 regulations differ from my coding by more than 1 unit. The correlation between the two sets of codings is 0.73, similar to the correlation between the random forest model prediction and my own codings. I, again, show the distribution of the coding differences by topic (Figure A5). Overall, hand-coding seems to have more negative codings for "Hukou," "local public goods," "work," and "other" category, and tend to have more positive codings for the "social insurance" type. Unfortunately, unlike the random forest model, it is harder to analysis and visualize the important factors in determining the result of the MLP model.

Figure A5: Difference between hand coding and MLP coding results, by topic



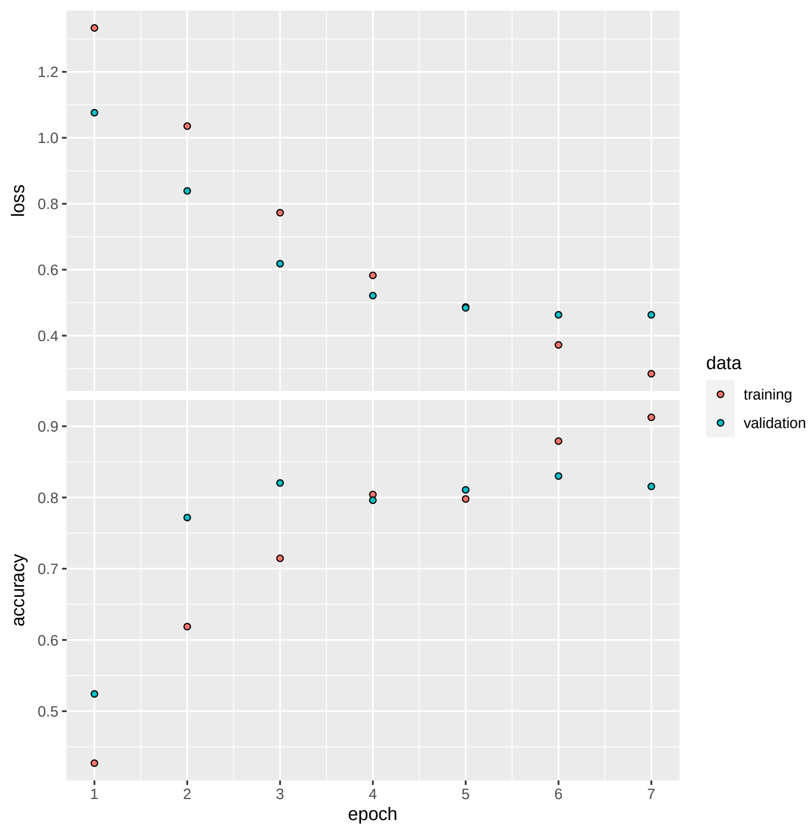
Note: This figure presents the comparison between the MLP coding and the manual coding, by showing distribution of the gap between the hand coding and MLP coding results by topic.

A.2.4 Method 3: Convolutional Neural Network with Word-Embedding

The third method is using convolutional neural network (CNN) with word embedding. First, as in the MLP model, top 2000 words are chosen to form the vocabulary. However, instead of an indicator variable for whether the word shows up in a regulation, words are represented as vectors using the word embedding method. Each dimension of the vector represent the feature of the word in a certain space. The entire regulation text can then be represented as an ordered collection of vectors, and this forms an embedding layer. Second, a convolutional layer is formed by applying an operator (filter) to the entirety of the text such that it transforms the information in the text. Third, the convolutional layer is flattened to a lower dimension to form a pooling layer, in order to avoid overfitting. Afterwards, similar to the MLP method, a fully connected hidden layer is used and an output layer will eventually generate the predictions. The word-embedding, filters, weights connecting different layers are computed to maximize the prediction accuracy using the training dataset.

Just to compare with the MLP network, the CNN with embedding has some advantages. First, instead of treating each individual word as a distinct signal, the word embedding use vectors to represent words, and words that are similar (for the purpose of eventual regulation classification task) should have vectors that are similar. Thus, more information is contained. Second, in the convolutional layer, not only the individual words, but also the adjacent words are used, again retaining more information in the text.[1]

Figure A6: CNN model results



Note: This figure shows the training process of the CNN model. The horizontal axis depicts the epoch, where one epoch is one round of model training using the regulations in the training set. The red dots represent the training data, and the green dots represent the validation data. The vertical axis in the top figure shows the value of the loss function, and the vertical axis in the bottom figure shows the value of accuracy. In this analysis, categorical crossentropy is used as the loss function. The smaller the loss, the better the model fit. Accuracy measures the opposite: the higher the accuracy, the better the model fit.

The parameter value of the model are as follows: 256 embedding dimensions (i.e., a word is represented using a 256 dimension vector), out of which 20% are dropped to avoid overfitting; 250 filters, each with a size of 3, and the filters move 1 word each time when scanning through the regulation text; 250 neurons in the hidden layer, again 20% dropped before feeding into the transfer function to generate the output layer.

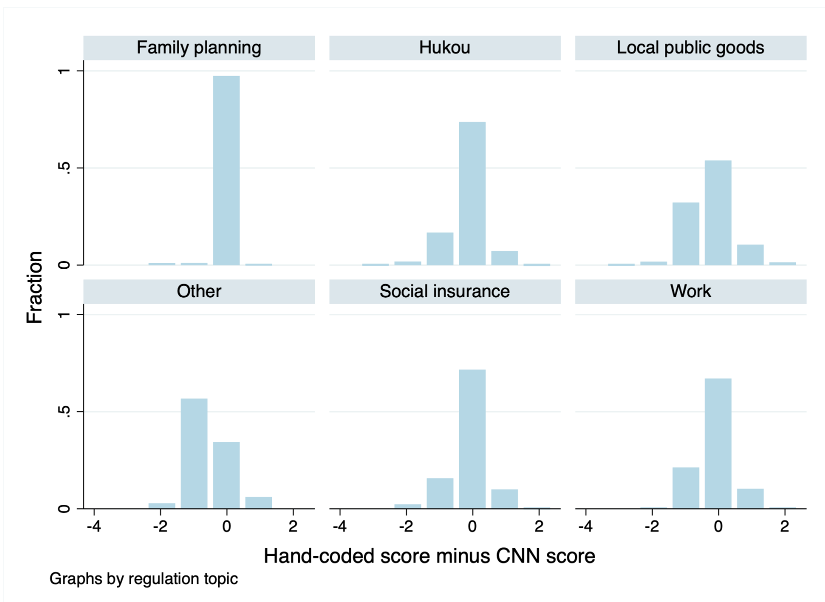
Similar to the MLP method, the model is trained with the 686 regulations with three codings, where 70% are used for training and 30% are for validation. The results of the training process are shown in Figure A6. Both the training accuracy and the validation accuracy increase over

[1]In the results shown below, the main text will only include the top 2000 words. However, increasing the size of the vocabulary (e.g., to include all words in the training text) does not significantly improve the prediction accuracy.

epochs, and the validation accuracy is around 81% in the last epoch. Compared the MLP result, the training accuracy and the validation accuracy are lower and improve substantially over time.

Among all the regulations, 2,887 regulations generate the same score as my own coding, 960 regulations have 1 unit larger coding, 340 with 1 unit smaller, and 86 regulations differ from my coding by more than 1 unit. The correlation between the two sets of codings is 0.71, slightly smaller than the ones using random forest and MLP. I, again, show the distribution of the coding differences by topic (Figure A7). Overall, hand-coding seems to have more negative codings for "Hukou," "local public goods," "work," and "other" category, and tend to have more positive codings for the "social insurance" type. This pattern is similar with the MLP ones.

Figure A7: Difference between hand coding and CNN coding results by topic



Note: This figure presents the comparison between the CNN coding and the manual coding, by showing distribution of the gap between the hand coding and CNN coding results by topic.

A.2.5 Summary of the Three NLP Methods

Table A3 presents a summary of the three NLP methods. The random forest method has an average of 20.70% out of bag error rate across 200 trees, and in terms of final prediction outcome in the training set, it performs the best (684 out of 686 correctly coded). The MLP method also approaches 100% prediction accuracy in the training regulation in the last epoch, and the validation accuracy is about 83%. Overall, 652 out of 686 are correctly coded. The CNN methods does not achieve the same level of training accuracy, but the validation accuracy is very similar to MLP. Overall, 624 out of 686 are correctly coded.

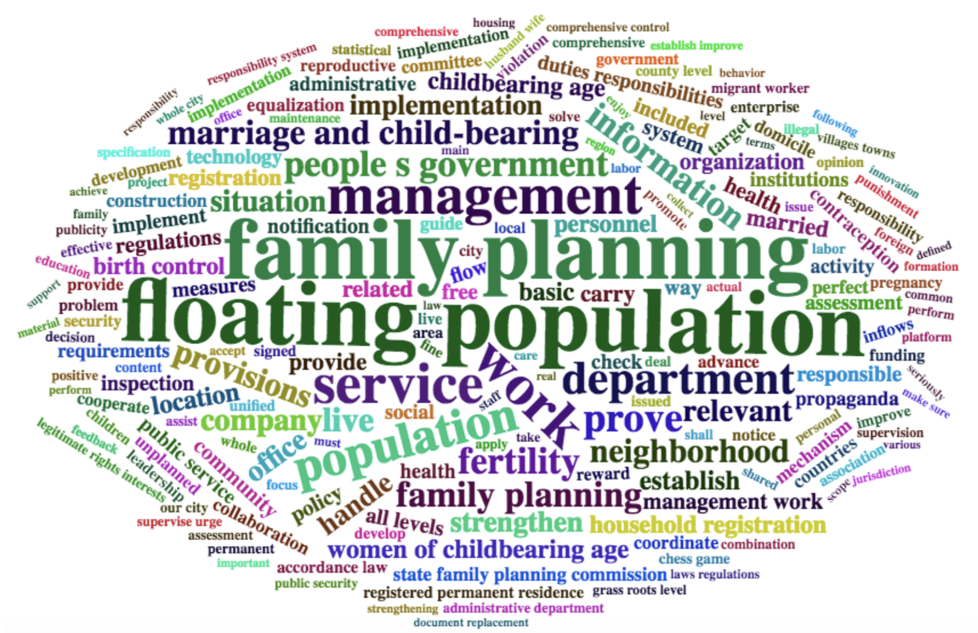
Table A3: Comparison of the three NLP methods

Method name	**Random forest**	**Multilayer perceptrons**	**CNN with word embedding**
Method description			
Method type	Decision tree	Neural network	Neural network
Elements of the text used	Words and word frequencies	Words and word frequencies	Words and word contexts
Training and validation			
Total training dataset	686	686	686
Training sample	Random draw with replacement	70% for training	70% for training
Validation sample	Not included in the random draw	30% for validation	30% for validation
Validation method	Average out-of-bag error rate	Validation accuracy	Validation accuracy
Validation result	20.70% out-of-bag error rate	83%	81%
Correctly coded	684 out of 686	652 out of 686	624 out of 686
Application to the full sample			
Correlation with manual coding, all 4,273 regulations	0.734	0.728	0.712

A.3 Summary of Statistics of the Regulation Data

Section 3 shows the wordclouds for the work-related and local-public-good-related regulation. Figures A8, A9, A10, and A11 show the corresponding wordclouds for regulations on family planning issues, Hukou-related administrative issues, social insurance issues, and other miscellaneous issues.

Figure A8: The most frequent words on the topic of family planning



Note: This figure shows the most frequent words of the 418 migrant-related regulations on family planning issues out of 4,273 regulations. A larger font indicates a higher frequency.

Figure A9: The most frequent words on the topic of Hukou

Note: This figure shows the most frequent words of the 1,247 migrant-related regulations on Hukou issues out of the 4,273 regulations. A larger font indicates a higher frequency.

Figure A10: The most frequent words in the topic of social insurance

Note: This figure shows the most frequent words of the 261 migrant-related regulations on social insurance issues. A larger font indicates a higher frequency.

Figure A11: The most frequent words on miscellaneous issues

Note: This figure shows the most frequent words of the 494 migrant-related regulations on miscellaneous issues out of the 4,273 regulations. A larger font indicates a higher frequency.

Table A4 shows that in the 2001–2007 period, 748 new regulations were enacted on migrant issues, with a mean score of 1, and 179 prefectures enacted at least one new regulation. In the 1995–2001 period, the numbers are much smaller: 144 new regulations in total and a mean score of 0.2. Forty-nine prefectures have some regulations, but only nineteen of them have positive regulations. Of the nineteen positive-regulation prefectures, only one has regulations about work-related issues, but all nineteen have administrative-related regulations. Among these nineteen prefectures, twelve are capital prefectures of provinces, with pro-migrant regulations about receiving local Hukou through purchase of commercial apartments and some specific issues on migrant workers.[2] There were only a few migrant-regulation changes before 2001, and they were concentrated in a few big prefectures on Hukou issues. In the 1978–1995 period, the mean regulation score is –0.1, and it is entirely driven by the Hukou related regulations.

[2]The twelve prefectures are Beijing, Changsha, Chongqing, Guangzhou, Huhehaote, Jinan, Shanghai, Shijiazhuang, Wuhan, Wulumuqi, Xi'an, and Zhengzhou. The other seven prefectures are Dalian, Huainan, Huizhou, Qingdao, Wuxi, Xiamen, and Xuzhou.

Table A4: Descriptive statistics on migrant-related regulations from 1978 to 2007

	# of regulations	Mean score	# of prefectures with			
			Any	(+)	0	(-)
2002–2007						
Total	748	1	179	161	84	8
Work	286	1.4	121	120	12	0
Local public goods	56	1.5	33	33	1	0
Social insurance	103	1.7	55	54	5	0
Family planning	55	0	35	0	35	0
Hukou	179	0.3	73	53	36	8
Miscellaneous	69	0.1	42	7	40	0
1994–2001						
Total	144	0.2	49	19	46	8
Work	1	1	1	1	0	0
Local public goods	0	0	0	0	0	0
Social insurance	2	1	1	1	1	0
Family planning	28	0	21	0	21	0
Hukou	110	0.2	44	19	39	8
Miscellaneous	3	0.3	1	1	1	0
1978–1995						
Total	69	-0.1	30	2	30	6
Work	0	0	0	0	0	0
Local public goods	4	0	2	0	2	0
Social Insurance	1	0	1	0	1	0
Family planning	16	0	14	0	14	0
Hukou	47	-0.1	25	2	25	6
Miscellaneous	1	0	1	0	1	0

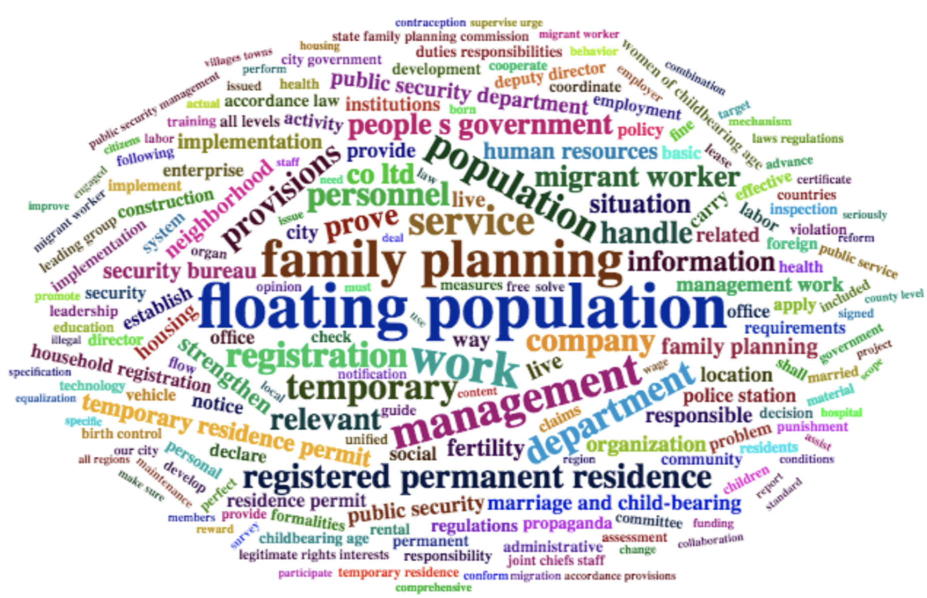
Note: This table presents the number of regulations, regulation scores, and the number of prefectures with positive, neutral, or negative regulations in three periods: 1978–1995, 1996–2001, and 2001–2007.

Figures A12, A13, and A14 show the most frequent words for regulations coded as positive, neutral, and negative, respectively.

Figure A12: The most frequent words in the regulations coded as positive

Note: This figure shows the most frequent in the regulations that are coded as positive, according to the manual coding. A larger font indicates a higher frequency.

Figure A13: The most frequent words in the regulations coded as neutral



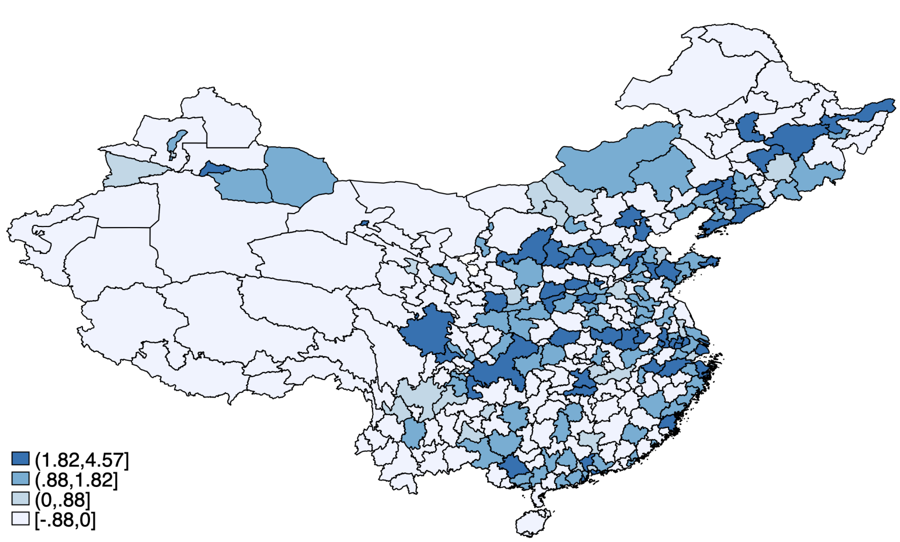
Note: This figure shows most frequent words in the regulations that are coded as neutral, according to the manual coding. A larger font indicates a higher frequency.

Figure A14: The most frequent words in the regulations coded as negative

Note: this figure shows the most frequent words in the regulations that are coded as negative, according to the manual coding. A larger font indicates a higher frequency.

Figure A15 shows the geographical distribution of the prefecture-level change in migrant-friendliness from 2001 to 2007. The prefecture-level migrant friendliness is constructed as the sum of regulation scores enacted, and then I do an inverse-hyperbolic-sine transformation of the total regulation score. Overall, coastal prefectures had more changes, but many inland prefectures also made substantial changes.

Figure A15: Geographic distribution of prefecture-level regulation changes (2001–2007)



Note: Each polygon is a prefecture. The regulation score is the sum over all prefecture-level regulations related to migrants. Then I do an inverse-hyperbolic-sine transformation of the total regulation score. Darker colors represent larger changes.

B Additional Information on Other Datasets

B.1 Trade Data and Measures

B.1.1 Industry Crosswalk, from 2-digit GB Code to 2-digit SIC Code

The industrial composition is from the 2000 Industrial Enterprises Survey, which is conducted on Chinese manufacturing firms with annual sales of more than 500 million RMB and includes basic firm information such as name and address, financial information on sales, export values, fixed capital, wage payment and total sales costs, and total employment.[3] There are 145,546 firms in 2000 with positive sales revenue and wage information, more than 10 employees, and a valid industry code. The industry code is the 4-digit Chinese Industry Code, which I aggregate to the 2-digit level. The 2-digit Chinese Industry Code is slightly finer than the 2-digit SIC code, with the crosswalk between the codes shown in Table B1. In addition, due to imperfect matching of the primary metal industry and the fabricated metal industry across the two sets of codes, I combine the two using export weights.

[3] The 1995 Industrial Enterprise Survey data is not available.

Table B1: Crosswalk, 2-digit Chinese industry code (GB) to 2-digit U.S. industry code (SIC), secondary sector

GB code	GB description	SIC	SIC description
6	Mining and washing of coal	12	Coal and lignite
7	Extraction of petroleum and natural gas	13	Crude petroleum and natural gas
8	Mining and processing of ferrous metal ores	10	Metallic ores and concentrates
9	Mining and processing of non-ferrous metal ores	10	Metallic ores and concentrates
10	Mining and processing of nonmetal ores	14	Nonmetallic minerals, except fuels
11	Mining of other ores	14	Nonmetallic minerals, except fuels
13	Processing of food from agricultural products	20	Food and kindred products
14	Manufacture of foods	20	Food and kindred products
15	Manufacture of liquor beverages and refined tea	20	Food and kindred products
16	Manufacture of tobacco	21	Tobacco manufactures
17	Manufacture of textile	22	Textile mill products
18	Manufacture of textile fabrics wearing apparel and accessories	23	Apparel and related products
19	Manufacture of leather, fur, feather and related products and footwear	31	Leather and leather products
20	Processing of timber, manufacture of wood, bamboo, rattan, palm and straw products	24	Lumber and wood products, except fuel
21	Manufacture of furniture	25	Furniture and fixtures
22	Manufacture of paper and paper products	26	Paper and allied products
23	Printing, production of recording media	27	Printing, publishing, and allied products
24	Manufacture of articles for culture, education, art, sport and entertainment activities	26*	Paper and allied products
25	Processing of petroleum and coking	29	Petroleum refining and related prod
26	Manufacture of raw chemical material and chemical products	28	Chemicals and allied products
27	Manufacture of medicines	28†	Chemicals and allied products
28	Manufacture of chemical fibers	28	Chemicals and allied products
29	Manufacture of rubber	30	Rubber and miscellaneous plastics products
30	Manufacture of plastics products	28‡	Chemicals and allied products
31	Manufacture of non-metallic mineral products	32	Stone, clay, glass, and concrete products
32	Smelting and pressing of ferrous metals	300⋇	Metal processing and products
33	Smelting and pressing of non-ferrous metals	300⋇	Metal processing and products
34	Manufacture of metal products	300⋇	Metal processing and products
35	Manufacture of general purpose machinery	35	Machinery, except electrical
36	Manufacture of special purpose machinery	35	Machinery, except electrical
37	Manufacture of transportation machinery	37	Transportation equipment
39	Manufacture of electrical machinery and equipment	36	Electrical machinery, equipment, and products
40	Manufacture of communication equipment, computers and other electric equipment	36	Electrical machinery, equipment, and products
41	Manufacture of measuring instruments and machinery	38	Scientific and professional instruments
42	Manufacture of artifacts and other manufacturing	39	Miscellaneous manufactured commodities
43	Recycling	91	Scrap and waste material

*https://www.osha.gov/pls/imis/sicsearch.html?p_sic=&p_search=stationery.
†https://www.osha.gov/pls/imis/sicsearch.html?p_sic=&p_search=drug.
‡https://www.osha.gov/pls/imis/sicsearch.html?p_sic=&p_search=plastic.
⋇https://www.osha.gov/pls/imis/sicsearch.html?p_sic=&p_search=metal. Here the SIC 300 will be the weighted average of SIC 33 (Primary metal products) and 34 (Fabricated metal products).

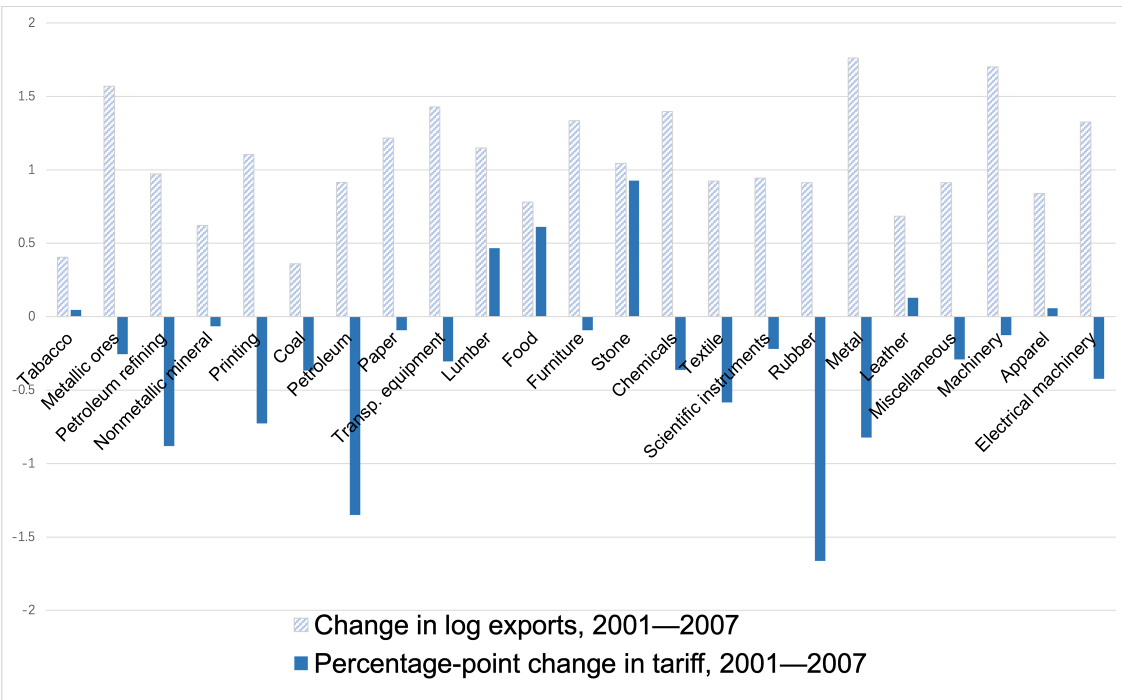
B.1.2 Changes in Tariffs

Tariff data is the applied tariff (AHS) at the 2-digit SIC level from the World Bank.[4] The tariff on Chinese exports is calculated as the weighted average of import tariff imposed by each destination country, with the 1995 export share as weights:

$$\tau_{jt}^{X} = \sum_{n} \frac{X_{j,1995}^{cn}}{\sum_{n'} X_{j,1995}^{cn'}} \tau_{jt}^{cn},$$

where $X_{j,1995}^{cn}$ is the Chinese exports to country n in industry j in 1995 and τ_{jt}^{cn} is the import tariff on Chinese exports to country n in industry j in year t.

Figure B1: Distribution of tariff changes and export growth across industries (2001–2007)



Note: Each bar is an industry. Horizontally sorted by value of exports in the industry in 2000.

Figure B1 plots the change in log exports (in light blue, shaded) and percentage-point changes in tariffs (in navy) in the 2001–2007 period. Industries at the 2-digit SIC level are sorted by the value of exports in the industry in 2000. We can see that changes in tariff levels varied greatly across industries.

Changes in tariffs on Chinese imports are calculated in a similar way as the changes in tariffs on Chinese exports:

$$\Delta \text{tariffs on imports}_{it} = \sum_{j} \beta_{ij} \left[\ln(1 + \tau_{jt'}^{M}) - \ln(1 + \tau_{jt}^{M}) \right],$$

[4]Source: `wits.worldbank.org`.

$$\text{where} \qquad \beta_{ij} = \frac{\lambda_{ij}\frac{1}{\theta_{ij}}}{\sum_{j'}\lambda_{ij'}\frac{1}{\theta_{ij'}}},$$

$\lambda_{ij} = \frac{L_{ij}}{L_i}$ is the fraction of regional labor allocated to industry j, and $1-\theta_{ij}$ is the cost share of labor in industry j. λ_{ij} and θ_{ij} are calculated from the 2000 Industrial Enterprises Survey data. τ_{jt}^M is tariff on imports in industry j in time t, and is directly taken from the World Bank Database.

Changes in tariffs on inputs is calculated similarly, by replacing τ_{jt}^M with τ_{ijt}^I. Tariffs on inputs in prefecture i and industry j in t is calculated as

$$\tau_{ijt}^I = \sum_{j'} \frac{\text{input}_{ij}^{j'}}{\sum_{j''}\text{input}_{ij}^{j''}} \ln(1+\tau_{j't'}^M)$$

I use the input-output table from the 2002 Chinese Regional Input-Output Table to calculate each industry's contribution to a certain industry ($\text{input}_{ij}^{j'}$) and to construct τ_{ijt}^I. The input-output table is available only on the province level; thus, my assumption here is that prefectures in the same province have the same input-output structure.

B.1.3 Alternative Trade Exposure Measures, as in Autor et al. [2013]

This section shows the construction of market-access-based trade shocks. The idea is this: suppose the overall export (import) increases in a certain industry over time at the national level, and per capita export growth can be calculated by dividing the increase in exports (imports) by the total number of people employed in the industry. Distributing the per capita export growth across regions according to the share of employment in the industry in a certain region, the overall regional trade shock is generated by summing over industries. Specifically, following Autor et al. [2013], the formula to calculate regional export exposure is as follows:

$$\Delta IPW_{it}^M = \sum_j \frac{L_{ijt}}{L_{jt}}\frac{\Delta M_{jt}}{L_{it}} = \sum_j \frac{L_{ijt}}{L_{it}}\frac{\Delta M_{jt}}{L_{jt}},$$

$$\Delta IPW_{it}^X = \sum_j \frac{L_{ijt}}{L_{jt}}\frac{\Delta X_{jt}}{L_{it}} = \sum_j \frac{L_{ijt}}{L_{it}}\frac{\Delta X_{jt}}{L_{jt}},$$

where L_{it} is the start-period employment (year t) in region i, L_{jt} is the start-period employment in industry j, L_{ijt} is the start-period employment in region i and industry j. ΔM_{jt} is the observed change in China's imports from the rest of the world in industry j between the start and the end of the period. The labor market exposure to import competition is the change in import exposure per worker in a region (in [Autor et al., 2013], it is the change in Chinese import exposure), where imports are apportioned to the region according to its share of national industry employment. Meanwhile, the export exposure is calculated by replacing the observed change in China's imports from the world ΔM_{jt} with China's exports to the world ΔX_{jt}.

The primary measure of interest is ΔIPW_{it}^X. The Bartik instrument uses the overall national growth to generate regional growth, by interacting with regional initial conditions. The benefit here is that it will be free of other local shocks that are correlated with local export growth. However, using the observed trade volume increase might still be problematic, since the overall trade increase might be correlated with overall economic growth, in which case the result would capture the "economic growth effect" instead of the "trade growth effect." Thus, I instrument the trade volume change further in two ways: the importing country's income growth and gravity dummies.

The GDP-based instrument is constructed as follows: suppose a country's fraction of income allocated to different industries' consumption (imports) does not change over time, and the fraction of imports in an industry that comes from China also does not change over time, then the growth of demand for Chinese goods will be from the growth of the importing country's income level. Specifically, the import value of Chinese goods in industry j (used as the superscript rather than as the subscript) and year t is constructed as follows:

$$\begin{aligned}
\mathrm{X}_{jt} &= \sum_{n} \frac{X^{cn}_{jt^*}}{\sum_{n'} X^{n'n}_{jt^*}} \frac{\sum_{n'} X^{n'n}_{jt^*}}{\sum_{j'} \sum_{n'} X^{n'n}_{j't^*}} GDP^n_t \\
&= \sum_{n} \frac{X^{cn}_{jt^*}}{X^{n}_{jt^*}} \frac{X^{n}_{jt^*}}{X^{n}_{t^*}} GDP^n_t \\
&= \sum_{n} \frac{X^{cn}_{jt^*}}{X^{n}_{t^*}} GDP^j_t,
\end{aligned}$$

where $\frac{X^{cn}_{jt^*}}{X^{n}_{jt^*}}$ is the fraction of imports in industry j and country n that comes from China in baseline year t^*, and $\frac{X^{n}_{jt^*}}{X^{n}_{t^*}}$ is the fraction of imports in industry j and country n out of the total import value. Then the export market access shock with the GDP measure in industry j between year t and t' is defined as

$$\Delta X^{GDP}_{jt} = \sum_{n} \frac{X^{cn}_{jt^*}}{X^{n}_{t^*}} \left(log(GDP^n_{t'}) - log(GDP^n_t) \right).$$

Alternatively, I use gravity dummies instead of GDP growth. First, I run a regression of log pairwise country imports on origin and destination country dummies, controlling for geographic distances. The export market access shock with the gravity measure is as follows:

$$\Delta X^{Gravity}_{jt} = \sum_{n} \frac{X^{cn}_{jt^*}}{X^{n}_{t^*}} \left(D_{n,t'} - D_{n,t} \right).$$

ΔX^{GDP}_{jt} and $\Delta X^{Gravity}_{jt}$ are used instead of ΔX_{jt} to calculate ΔIPW^X_{it}.

I also add a measure for the intermediate-goods market access shock:

$$\Delta IPW^I_{it} = \sum_{j} \frac{L_{ijt}}{L_{it}} \hat{I}_{ijt}$$

and

$$\hat{I}_{ijt} = \sum_{j'} \frac{input^{ij}_{j'}}{\sum_{j''} input^{ij}_{j''}} \left[\frac{M_{j't'}}{L_{j't}} - \frac{M_{j't}}{L_{j't}} \right].$$

B.2 Data on Migrants

The information about the number of migrants is from the 2000 and 2010 censuses and the 2005 1% population survey. The 2000 individual data is a 0.1% random sample of the population, and the 2005 data is a 0.2% random sample of the population. I use 2010 aggregate prefecture-level data for the analysis since the individual data is not available.[5]

A person is defined as a migrant if he or she has been living in a place other than the Hukou registration place for more than six months or has left the Hukou registration location for more than six months. There were 144 million migrants in 2000, 0.39 million per prefecture. In 2010, the number increased to 261 million, 0.77 million per prefecture.

Table B2: Summary of statistics of census data, 2000, 2005, and 2010, all 340 prefectures

Mean (s.d.) in million persons	2000	2005	2010
Total population	3.5 (2.7)	3.8 (4.0)	3.9 (3.2)
# of locals	3.1 (2.3)	3.2 (3.3)	3.1 (2.3)
# of migrants	.39 (.63)	.56 (1.20)	.77 (1.31)
By migration distance			
within prefectures (short distance)	.19 (.22)	.25 (.45)	.11 (.28)
across prefectures (medium distance)	.09 (.15)	.08 (.22)	.39 (.41)
across provinces (long distance)	.11 (.37)	.23 (.88)	.25 (.85)
By reason of migration			
Work	.12 (.36)	.26 (.83)	.41 (.80)
Family	.04 (.06)	.09 (.18)	.13 (.15)
Marriage	.01 (.02)	.04 (.08)	.04 (.04)
Other	.20 (.26)	.16 (.38)	.15 (.24)
By years of education			
<=12 years of education	.34 (.55)	.45 (1.11)	.59 (0.99)
> 12 years of education	.03 (.07)	.06 (.23)	.13 (.29)
By years since moved here			
<=3 years	.19 (.40)	.25 (.69)	.43 (.70)
> 3years	.16 (.22)	.26 (.68)	.32 (.62)

There is also information about how far the person migrated. A migrant is defined as a within-prefecture migrant if the Hukou prefecture and the residence prefecture are the same. Between-prefecture within-province migration means the Hukou prefecture and the residence prefecture are different but in the same province. A between-province migrant is one whose Hukou province and residence province are different. The total number of migrants is decomposed into these three categories to see if changes in tariffs on exports and regulation changes affected them differently.

[5]The census and 1% population survey are conducted via personal visits. To address potential issues related to under-reporting, the Census Bureau randomly samples some neighborhoods after the census concludes and check the nonresponse rate. The nonresponse rate in the 2000 census is 1.81%. Source: `www.stats.gov.cn/tjsj/ndsj/renkoupucha/2000pucha/html/append21.htm`.

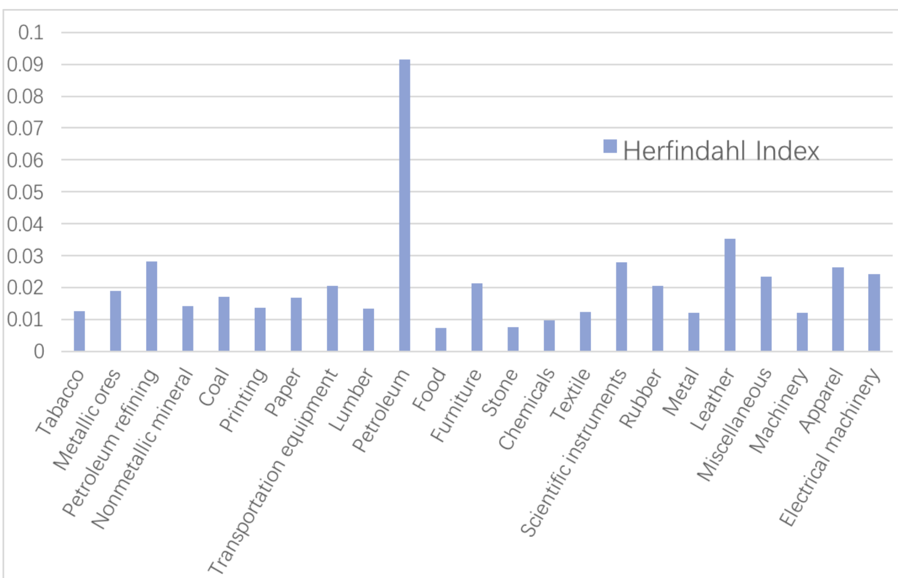
When I study the effect of the 2001–2007 changes in tariffs on exports on migrant flows, I use the 2000 and 2010 data, because it could have taken time for the regulations to affect actual migrant flows. In Section C.11.2, I exploit the timing of the regulation change and migrant flows to show whether the regulation drives the migrant flows or the other way around, and I use all three years of data.

The 2005 Population Survey contains a wealth of information on respondents.[6] For example, the respondents were asked about their medical insurance, pension, unemployment insurance, terms of contract, and wages. I use the 2005 social insurance measures to check whether in places with more pro-migrant regulations, migrants enjoy more social insurance and are paid higher wages. Also, industries are identified by two-digit SIC codes. The industry classification helps to construct the industry-level migrant share of total employment, i.e., the industry-level migrant intensity. In the manufacturing sector, manufacturing of communication equipment, computers, and other electronic equipment has 68% of migrant employment; mining and processing of ferrous metal ores has only 15%.

B.3 Supporting Evidence on Prefecture Sample Restrictions

The Herfindahl index by industry is shown in Figure B2. The petroleum industry has the highest level of concentration across prefectures.

Figure B2: Industry concentration across prefectures in 2000, Herfindahl Index



Note: Each bar is a industry, horizontally sorted by the value of exports in the industry in 2000.

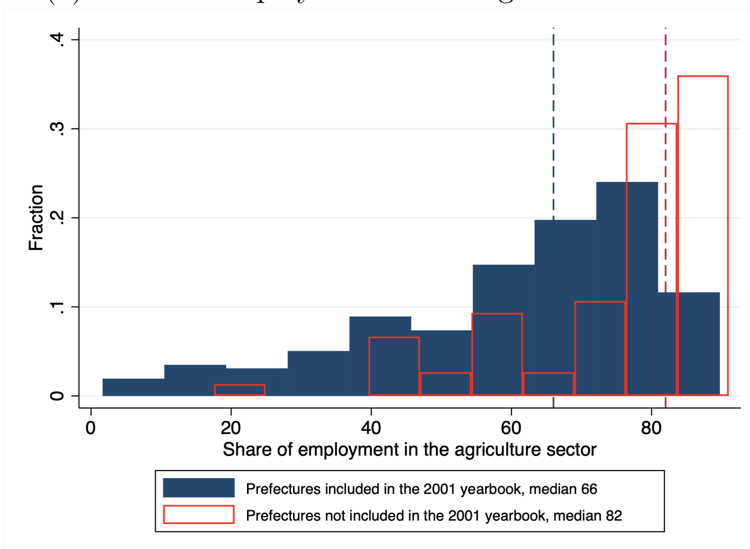
The distribution of agriculture employment share and urban population share, for prefectures included and not included in the 2001 Prefecture Statistic Yearbook is shown in Figure B3. Both measures are calculated using the 2000 population census data. Using both measures, it is clear

[6] The 2000 Census also has the industry and occupation information, but the coding is not standard GB code. There is no information on social insurance or wages in the 2000 sample.

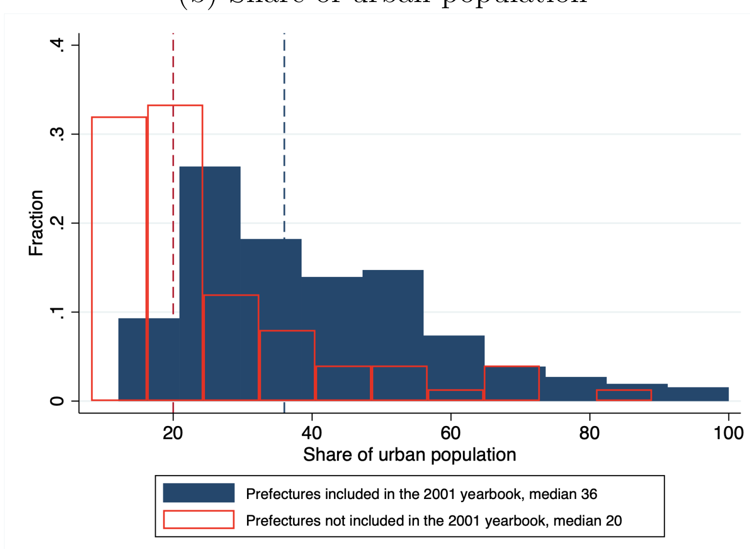
that prefectures that are included in the Prefecture Statistics Yearbook are more urban compared to prefectures that are not included in the Yearbook.

Figure B3: The distribution of agriculture employment share and urban population share, for prefectures included and not included in the 2001 Prefecture Statistic Yearbook

(a) Share of employment in the agriculture sector



(b) Share of urban population



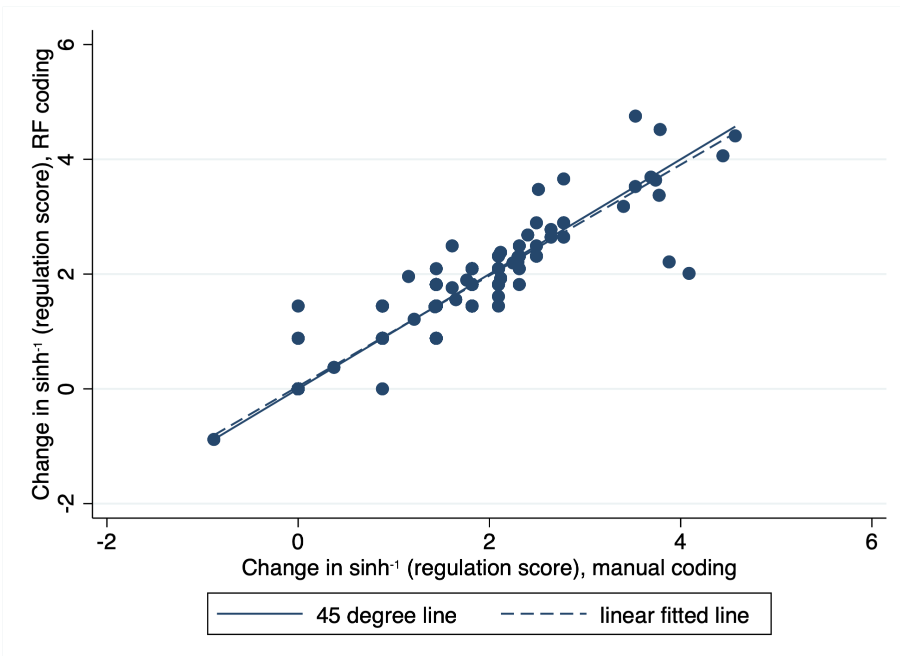
Note: The share of agriculture employment and the share of urban population is from the 2000 population census. The blue bars (lines) are for the prefectures that are included in the 2001 Prefecture Statistics Yearbook, and the red bars (lines) are for the prefectures that are not included in the 2001 Prefecture Statistics Yearbook.

C Additional Empirical Results

C.1 The Correlation between the Manually Coded Regulation Scores and the NLP Scores

Figure C1 shows the correlation between the change in the inverse-hyperbolic-sine-transformed regulation scores from 2001 to 2007 that is coded manually and the one coded using the random forest method. The correlation is 0.96 and the dots are very close to the 45 degree line.

Figure C1: The prefecture-level correlation between the change in regulation scores coded manually and coded by the random forest method (2001–2007), 333 prefectures

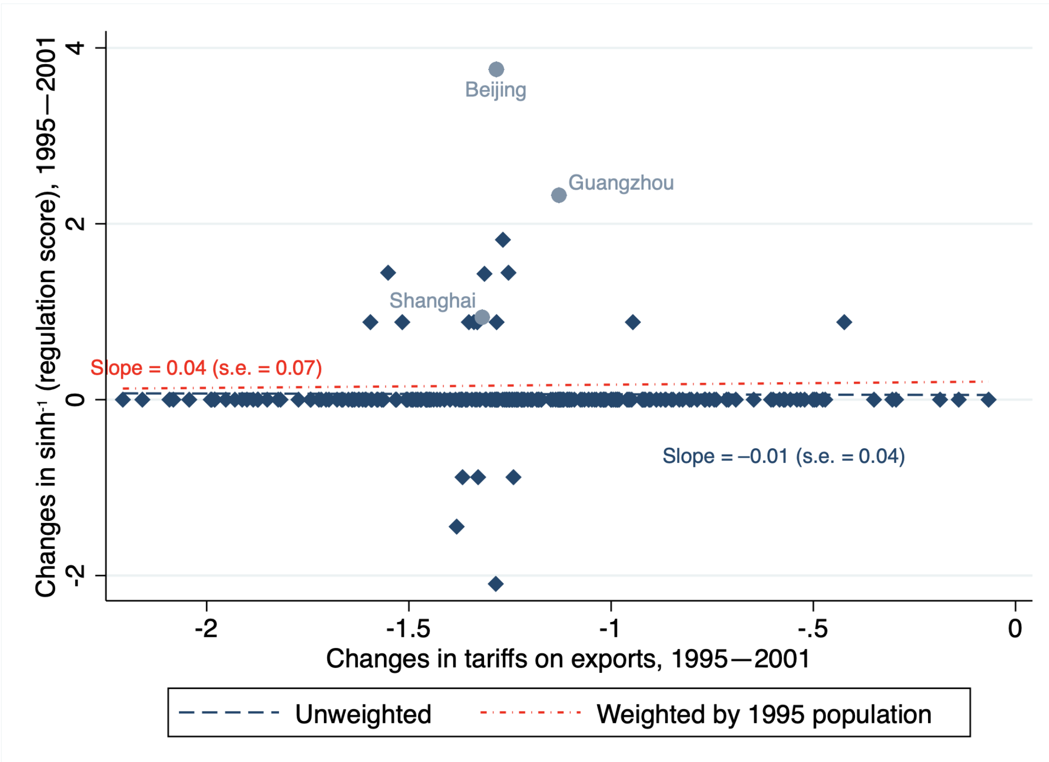


Note: Each dot is a prefecture. The horizontal axis is the change in the inverse-hyperbolic-sine-transformed regulation score from 2001 to 2007 using the manual coding, and the vertical axis is the change using the random forest method.

C.2 The Relationship between Changes in Tariffs on Exports and Changes in Regulation, 1995 to 2001

Figure C2 plots the relationship between changes in the inverse-hyperbolic-sine-transformed regulation score and changes in tariffs on exports in the 1995–2001 period to compare with Figure 5. Pre WTO accession, there were clearly few changes in migrant-related regulations (with insignificant coefficients of –0.01 to 0.04, while the coefficients are –1.84 to –0.90 and statistically significant in the post-WTO period), and the few prefectures that changed migrant regulations were provincial capitals. This reinforces the argument about the significance of the WTO effect.

Figure C2: Effect of changes in tariffs on exports on regulation change (1995–2001), 235 prefectures

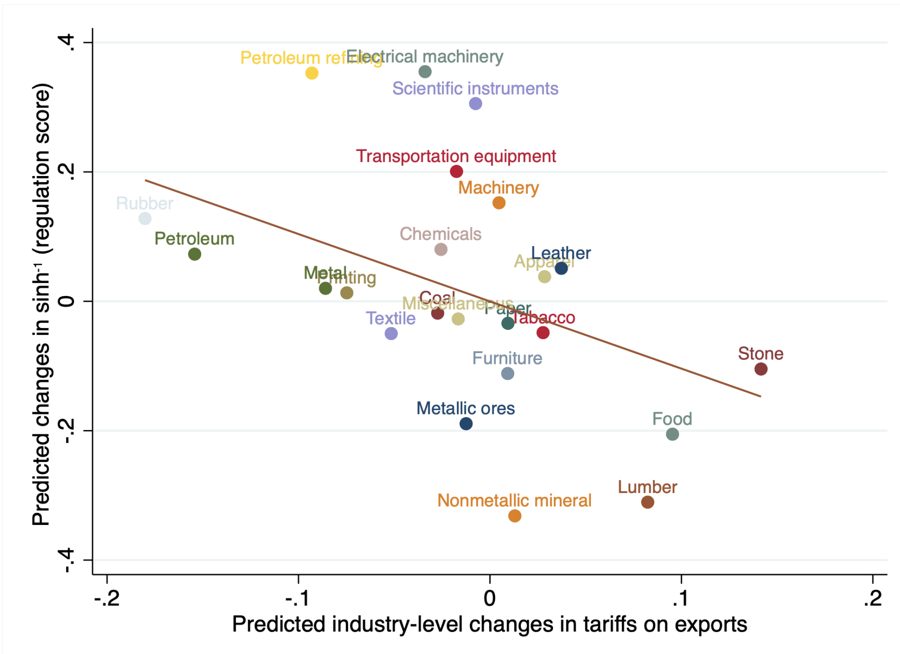


Note: Each dot is a prefecture. This figure presents the relationship between the change in tariffs on exports and the change in $\sinh^{-1}$(regulation score) in the 1995–2001 period. The dashed line presents the linear fitted line, unweighted, and the dotted line presents the linear fitted line, weighted by the 1995 population.

C.3 The Relationship between Changes in Tariffs on Exports and Changes in Regulation at the Industry Level

Figure C3 plots the relationship between changes in the inverse-hyperbolic-sine-transformed regulation score and predicted changes in tariffs on exports in the 2001–2007 period at the industry level, following Borusyak et al. [2018]. The industry weighted IV regression generates the same point estimate of –1.04 as in Table 1 Column (1).

Figure C3: Effect of changes in tariffs on exports on regulation changes (2001–2007), 23 industries



Note: Each dot is an industry. This figure presents the relationship between changes in tariffs on exports and the the change in $\sinh^{-1}$(regulation score), both converted to the industry level.

C.4 Main Results with Alternative NLP Measures

Instead of using the random forest codings for regulations, I use the average regulation codings by the three NLP methods, and the MLP and CNN codings in Table C1. The results are similar to the ones in Table 1.

Table C1: Effects of changes in tariffs on exports on changes in regulation scores (2001–2007), using the average of the three NLP measures, and MLP and CNN separately

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A.	Dependent variable: Δ $\sinh^{-1}$(regulation score), 2001–2007, average of NLP codings						
Δ tariffs on exports, 2001–2007	-1.00***	-0.89***	-1.38***	-1.11***	-1.47***	-1.27**	-0.97*
	(0.30)	(0.25)	(0.32)	(0.29)	(0.50)	(0.47)	(0.52)
Observations	333	333	317	317	248	248	235
R-squared	0.11	0.16	0.12	0.17	0.09	0.13	0.16
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel B.	Dependent variable: Δ $\sinh^{-1}$(regulation score), 2001–2007, MLP coding						
Δ tariffs on exports, 2001–2007	-1.06***	-0.94***	-1.44***	-1.16***	-1.53***	-1.34***	-1.00*
	(0.30)	(0.26)	(0.33)	(0.29)	(0.50)	(0.48)	(0.52)
Observations	333	333	317	317	248	248	235
R-squared	0.09	0.15	0.11	0.16	0.08	0.12	0.15
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel C.	Dependent variable: Δ $\sinh^{-1}$(regulation score), 2001–2007, CNN coding						
Δ tariffs on exports, 2001–2007	-0.92***	-0.83***	-1.27***	-1.06***	-1.37***	-1.21**	-0.94*
	(0.28)	(0.25)	(0.30)	(0.29)	(0.48)	(0.47)	(0.51)
Observations	333	333	317	317	248	248	235
R-squared	0.14	0.18	0.15	0.18	0.11	0.15	0.17
Mean (sd) $\Delta\sinh^{-1}$(regulation score), average	0.89 (1.02)		0.89 (1.02)		1.03 (1.05)		1.06 (1.06)
Mean (sd) $\Delta\sinh^{-1}$(regulation score), MLP	0.90 (1.05)		0.90 (1.04)		1.03 (1.08)		1.06 (1.08)
Mean (sd) $\Delta\sinh^{-1}$(regulation score), CNN	0.90 (1.01)		0.90 (1.01)		1.03 (1.04)		1.06 (1.04)
Mean (sd) Δ tariffs on exports	-0.16 (0.20)		-0.14 (0.18)		-0.18 (0.15)		-0.18 (0.15)

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. In Panel (a), the dependent variable is the change in the inverse-hyperbolic-sine-transformed average regulation score using the three NLP methods. Panel (b) and Panel (c) use the MLP coding and the CNN coding, respectively. The sample restrictions are the same as in Table 1, and all three panels have the same controls as in Table 1 Panel A.

C.5 Main Results with Panel Specifications

The main specification in Section 6.2 uses the long differences between 2001 and 2007, and control for the 1995 to 2001 changes. Alternatively, the two long-differences can be stacked together:

$$\Delta\text{sinh}^{-1}(\text{regulation score})_{it} = \alpha_0 + \alpha_1\Delta \text{T}_{it} + \alpha_2\Delta \text{T}_{it} \times I_t + X_{it}\Gamma + I_i + I_t + \epsilon_{it},$$

where $t \in \{1995–2001, 2001–2007\}$, i represents a prefecture, ΔT_{it} is the change in tariffs on exports, I_i is prefecture fixed effects and I_t is a dummy which equal to 1 if it is the second period (2001–2007). Note that I allow for differential impact of changes in tariffs on exports on regulation changes in the first and the second period, since as described in Section 2, prefecture-level governments were granted discretion on migrant-related issues mostly in the second period.

The results are shown in Table C2. Column (1) regresses the change in the inverse-hyperbolic-sine-transformed regulation score on the changes in tariffs on exports, changes in tariffs on imports, and changes in tariffs on inputs, with each tariff change interacted with the period dummy, controlling for prefecture fixed effects and the period dummy. The coefficient estimates for the tariff changes are insignificant, indicating that in the 1995–2001 period, trade did not drive changes in migrant regulations. The coefficient for the changes in tariffs on exports interacted with the 2001–2007 period dummy is negative and statistically significant, indicating that the change in tariffs on exports after the WTO accession significantly affected the migrant friendliness. Columns (2)–(4) add additional controls for prefecture characteristics at the beginning of each period, and the results are similar to Column (1). Columns (5)–(8) use random forest coding stead of manual coding, and the results are similar.

Table C2: The relationship between changes in tariffs on exports changes in regulation scores, panel specification

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Δ sinh^{-1}(regulation score), manual				Δ sinh^{-1}(regulation score), random forest			
Δ tariffs on exports	0.28	0.27	0.25	0.25	0.28	0.27	0.25	0.25
	(0.23)	(0.23)	(0.24)	(0.24)	(0.24)	(0.23)	(0.23)	(0.23)
Δ tariffs on exports $\times$ I(01–07)	-1.66***	-1.42**	-1.31**	-1.30**	-1.53**	-1.26**	-1.20*	-1.20*
	(0.58)	(0.59)	(0.57)	(0.58)	(0.61)	(0.61)	(0.59)	(0.60)
	(0.58)	(0.59)	(0.57)	(0.58)	(0.61)	(0.61)	(0.59)	(0.60)
Δ tariffs on imports	-0.01	-0.02	-0.02	-0.02	-0.02	-0.02	-0.03	-0.03
	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.03)
Δ tariffs on imports $\times$ I(01–07)	-0.08	-0.08	-0.07	-0.07	-0.08	-0.08	-0.08	-0.08
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Δ tariffs on inputs	0.23*	0.14	0.15	0.15	0.28*	0.18	0.18	0.18
	(0.13)	(0.11)	(0.13)	(0.15)	(0.14)	(0.12)	(0.14)	(0.15)
Δ tariffs on inputs $\times$ I(01–07)	-0.36**	-0.26	-0.30	-0.30	-0.33*	-0.21	-0.24	-0.24
	(0.17)	(0.17)	(0.19)	(0.19)	(0.18)	(0.18)	(0.20)	(0.20)
Observations	470	470	470	470	470	470	470	470
R-squared	0.71	0.72	0.72	0.72	0.70	0.72	0.72	0.72

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. Columns (1)–(4) use the change in the inverse-hyperbolic-sine-transformed regulation score by manual coding, and Columns (5)–(8) use random forest coding. All columns control for period fixed effects, prefecture fixed effects, and the initial regulation score. Columns (2) and (6) adds the log GDP per capita at the beginning of the period, Columns (3) and (7) adds the initial log wage, and Columns (4) and (8) adds the initial log population. The mean (s.d.) of Δ sinh^{-1}(regulation score), 2001–2007 is 1.03 (1.08) for manual coding and 1.04 (1.07) for random forest coding. The mean (s.d.) of Δ sinh^{-1}(regulation score), 1995–2001 is 0.06 (0.44) for manual coding and 0.05 (0.33) for random forest coding. The mean (s.d.) of changes in tariffs on exports, 2001–2007 is –0.18 (0.15), 1995–2001 is –1.22 (0.40). The sample restriction is the same as in Table 1 Column (7), with 235 prefectures per period.

Overall, the findings here are similar to the ones in Table 1.

C.6 Main Results with Score Levels

Instead of using inverse-hyperbolic-sine-transformed regulation scores, I use the level of scores in Table C3. The results are similar to the ones in Table 1.

Table C3: The effects of changes in tariffs on exports on changes in regulation score, using regulation score levels instead of inverse-hyperbolic-sine transformations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A.	Dependent variable: Δ regulation score, level, 2001–2007, manual coding						
Δ tariffs on exports, 2001–2007	-3.26***	-2.65***	-4.92***	-3.86***	-5.93***	-4.84***	-3.45*
	(1.03)	(0.83)	(1.07)	(0.93)	(1.72)	(1.69)	(1.85)
Δ tariffs on imports, 2001–2007	-0.24*	-0.27*	-0.29**	-0.36**	-0.52***	-0.56**	-0.53*
	(0.14)	(0.15)	(0.14)	(0.16)	(0.17)	(0.26)	(0.28)
Δ tariffs on inputs, 2001–2007	-0.72	1.49	-0.72	1.33	0.03	1.87*	1.60
	(0.58)	(0.88)	(0.55)	(0.89)	(0.59)	(1.00)	(1.11)
Regulation score, level, 2001	3.32**	3.26**	3.29**	3.26**	3.23**	3.23**	3.16**
	(1.30)	(1.24)	(1.29)	(1.25)	(1.26)	(1.24)	(1.17)
PNTR shock, 2001–2007		12.39*		8.17		4.45	-1.21
		(6.99)		(5.83)		(6.98)	(7.48)
MFA shock, 2001–2007		30.15***		28.27***		26.20***	26.54***
		(9.32)		(8.84)		(8.53)	(8.85)
Δ tariffs on exports, 1995–2001		0.55		0.51		0.41	0.37
		(0.66)		(0.62)		(0.92)	(0.99)
Δ tariffs on imports, 1995–2001		-0.06		0.00		0.02	0.05
		(0.07)		(0.06)		(0.09)	(0.09)
Δ tariffs on inputs, 1995–2001		-1.61***		-1.66***		-1.65***	
-1.70***		-1.81***	-1.61**				
		(0.52)		(0.51)		(0.58)	(0.69)
Δ log wage, 1995–2001							6.10**
							(2.22)
Δ log GDP p.c., 1995–2001							1.69*
							(0.94)
Observations	333	333	317	317	248	248	235
R-squared	0.23	0.28	0.25	0.29	0.25	0.28	0.32
Bootstrap p-value, Δ tariffs on exports	.002	.002	0	.003	.003	.023	.086
AKM1 p-value, Δ tariffs on exports	.029	.003	.004	.002	.022	.023	.048
AKM0 p-value, Δ tariffs on exports	.061	.035	.037	.045	.057	.059	.041
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel B.	Dependent variable: Δ regulation score, level, 2001–2007, RF coding						
Δ tariffs on exports, 2001–2007	-3.33***	-2.70***	-4.76***	-3.61***	-5.93***	-4.79**	-3.95**
	(1.11)	(0.91)	(1.26)	(0.99)	(1.96)	(1.83)	(1.90)
Observations	333	333	317	317	248	248	235
R-squared	0.37	0.42	0.42	0.45	0.42	0.44	0.46
Mean (sd) Δ regulation score, manual	2.20 (4.62)		2.20 (4.58)		2.66 (5.05)		2.77 (5.16)
Mean (sd) Δ regulation score, RF	2.32 (5.33)		2.33 (5.36)		2.82 (5.94)		2.93 (6.07)
Mean (sd) Δ tariffs on exports	-0.16 (0.20)		-0.14 (0.18)		-0.18 (0.15)		-0.18 (0.15)

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. This table replicates Table 1 by using the level of the regulation score instead of the inverse-hyperbolic-sine-transformed score.

C.7 Alternative Measure of Regulation Changes

In the main specification, I use the regulation score on a -2 to 2 scale, with -2 as the least migrant-friendly and 2 as the most migrant-friendly. In Table C4, I use a "negative (–1), neutral (0), and positive (+1)" scale (Column 2) and also a simple count of the number of regulations (Column 3) to check the robustness of the result . Then, I investigate the effect of tariff changes on changes in the regulation score by topic in Columns (4)–(8).

Table C4: The effects of changes in tariffs on exports on changes in the regulation score, using alternative measure of regulation changes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Δ sinh^{-1}(regulation score)		Δ sinh^{-1}(number		Δ sinh^{-1}(regulation score)			
2001–2007	5 levels	3 levels	of regulations)	Work	Local public good	Hukou	Social insurance	Other
Δ tariffs on exports	-1.60***	-1.47***	-0.68*	-1.24***	-0.74***	-0.39**	-0.28	-0.09
2001–2007	(0.50)	(0.41)	(0.35)	(0.41)	(0.22)	(0.19)	(0.36)	(0.06)
Δ tariffs on imports	-0.11**	-0.10**	-0.03	-0.07*	-0.04	-0.02	-0.08***	-0.01*
2001–2007	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.02)	(0.02)	(0.00)
Δ tariffs on inputs	-0.31**	-0.30**	-0.29**	-0.14	-0.11*	-0.21***	0.17	-0.06
2001–2007	(0.14)	(0.12)	(0.11)	(0.14)	(0.06)	(0.07)	(0.11)	(0.04)
Y, 2001	-0.12	0.10	0.15***	2.45***		-0.01	1.05***	-0.02
2001	(0.22)	(0.18)	(0.05)	(0.13)		(0.10)	(0.05)	(0.02)
Observations	248	248	248	248	248	248	248	248
R-squared	0.08	0.10	0.09	0.06	0.05	0.07	0.05	0.03
Mean (s.d.) of Y	1.00 (1.08)	.78 (.88)	.75 (.74)	.71 (.94)	.19 (.54)	.14 (.42)	.33 (.70)	.03 (.16)

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. This table replicates Table 1 Column (5), by changing the measure of regulation scores. Column (1) is the same as Table 1 Column (5), Column (2) uses three levels of regulation index {–1, 0, 1} instead of five levels, Column (3) uses the inverse-hyperbolic-sine-transformed number of regulations instead of regulation scores, and Columns (4)–(8) split the regulations by topic. The mean value (s.d.) of changes in tariffs on exports, 2001–2007 is –0.18 (0.15).

The results show that the effect of changes in tariffs on exports on regulation changes is robust to variation in the regulation measure. The five-level coding is the most informative about the migrant-friendliness, and the effect of changes in tariffs on exports is also the biggest and most significant among the first three columns. In the latter five columns, the effects of changes in tariffs on exports on work-related and local-public-good-related regulations are the largest and most significant. Overall, all columns are consistent with the main result.

C.8 Alternative Measure of Bartik Trade Exposures

To check the robustness of the main results with respect to the measure of tariff changes, I use industry labor shares as weights directly: $\beta_{ij}^{'} = \lambda_{ij}$ in Table C5 Column (2). Compared to Column (1), which replicates Table 1 Column (5), the coefficient on the changes in tariffs on exports is very similar to the main result.

Alternatively, I follow the Autor et al. [2013] measure of local-labor-market trade shocks and construct local-market-access shocks. The market-access shock is also a Bartik-style measure, with industry-level export growth distributed across regions, weighted by local-industry labor shares. The difference with the Autor et al. [2013] measure is that I use export growth instead of import growth, since export growth is more relevant in the Chinese context. Also, since Autor et al. [2013] analyze the effect of exposure to Chinese exports on the U.S. economy, the authors use Chinese exports to other developed countries as an instrument to capture the Chinese productivity growth effect. In my case, I want to capture the demand-side forces that led to the expansion of Chinese exports, so I use the GDP growth of the importing countries as an instrument. An alternative measure would be the change in country dummies from a bilateral trade gravity regression.

Table C5: The effects of trade on changes in regulation scores, using alternative measures of trade exposures

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent variable: Δ $\sinh^{-1}$(regulation score), 2001–2007						
Trade exposure measure:	Tariff-based		Market-access-based				
	Main	Labor share	OLS	OLS	OLS	GDP IV	Gravity IV
Export exposure, 2001–2007	-1.60***	-1.48***	0.04***	0.02**	0.03**	0.03**	0.03*
	(0.50)	(0.46)	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)
Import exposure, 2001–2007	-0.11**	-0.10**		0.02	0.01	0.00	0.01
	(0.04)	(0.04)		(0.02)	(0.02)	(0.02)	(0.02)
Input exposure, 2001–2007	-0.31**	-0.31**		0.17	0.14	0.12	0.13
	(0.14)	(0.14)		(0.11)	(0.12)	(0.12)	(0.12)
$\sinh^{-1}$(regulation score), 2001	-0.12	-0.12			-0.18	-0.19	-0.18
	(0.22)	(0.22)			(0.18)	(0.17)	(0.17)
Urban share, 2001					0.02***	0.02***	0.02***
					(0.01)	(0.01)	(0.01)
Observations	248	248	248	248	248	248	248
R-squared	0.08	0.07	0.07	0.09	0.16	0.16	0.16
Mean (s.d) of export shock	-0.18 (0.15)	-0.17 (0.15)	16.04 (6.50)				
First-stage F-stat	-	-	-	-	-	320	490

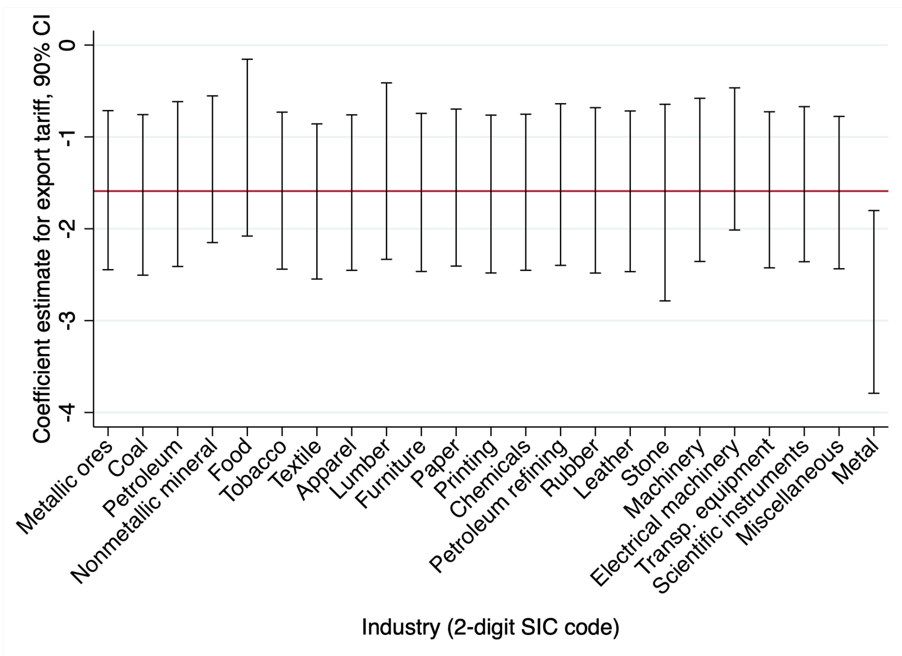
Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. This table replicates Table 1 Column (5), by changing the measure of trade exposure. The first two columns use changes in tariffs as the measures for trade exposure. Column (1) is the same as Table 1 Column (5), Column (2) uses the employment share weights instead of employment shares combined with labor intensity. Columns (3)–(7) use market-access-based measures for trade exposure, as in Autor et al. [2013]. Columns (3)–(5) are OLS estimates, Column (6) uses an GDP-based instrument variable, and Column (7) use an gravity-based instrument variable. The mean value (s.d.) of Δ $\sinh^{-1}$(regulation scores), 2001–2007 is 1.00 (1.08).

Table C5 Columns (3)–(7) show the results with the market-access-based shocks. Column (3) contains only the export shocks, Column (4) adds the import and input shocks, and Column (5) adds urban share of the prefecture as a control. Column (6) instruments the export shock with the GDP-based instrument. Column (7) uses the gravity-dummy-based instrument. The size of the coefficient on the export shock is robust across these specifications, but the IV coefficients are less significant. The results show that a $1,000 per worker increase in exports led to a 0.03-unit increase in the inverse-hyperbolic-sine-transformed regulation score. Again, I divide prefectures into big-, medium- and small-shock ones, and the difference in export shocks between the big- and small-shock ones is $14,000 per worker. This translates into a 0.42 higher increase in the inverse-hyperbolic-sine-transformed regulation score, which is comparable to the 0.42 difference found in the main regression with tariff changes (Table 1 Column 1).[7]

C.9 Adding Industrial Composition Controls

The regional tariff changes are generated using the interaction of prefecture-level industrial composition and industry-level tariff reductions. If certain industries drive variation and are correlated with other local factors that affect regulation changes directly, then the estimates for the effects of regional tariff changes would be biased. To check whether such an industry exists, I add industry employment shares one at a time and run the regression in Table 1 Column (5).

Figure C4: Coefficients from the main regression by adding employment shares by industry one by one



Note: Each bar is the 90% confidence interval of the coefficient estimate of changes in tariffs on exports from a regression as in Table 1 Column (5), controlling for a specific-industry share of total employment. The horizontal bar is the point estimate of -1.60 from Table 1 Column (5).

Figure C4 plots the coefficient estimates with 90% confidence intervals, and each bar is from a regression, including a specific-industry employment share. The coefficient estimates are relatively stable around –1.60, which is the estimate in Table 1 Column (5). Thus, the results are not sensitive to specific-industry effects.[8]

[7]The per capita export was about \$300 in 2001 and \$1,000 in 2007. The number of employed workers in the Industrial Enterprises Survey in 2000 is 50 million. Thus, the \$14,000 per worker difference is equivalent to \$580 per person and is comparable to the \$700 mean increase from 2001 to 2007.

[8]Including the metal industry employment share makes the coefficient on changes in tariffs on exports more negative, while the metal employment share itself has a significant negative effect.

C.10 Competition between Prefectures in Regulation Changes

Prefecture i's regulation change and tariff changes can affect not only its own regulation but also that of other prefectures. The most direct measure of the intensity of competition is to focus on nearby prefectures. Table C6 Column (1) replicates the result in Table 1 Column (5). Columns (2)–(4) consider the competition with other prefectures in the same province. Column (2) adds changes in tariffs, Column (3) adds regulation changes, and Column (4) controls for both. Columns (5)–(7) repeat the exercise by considering the competition with five nearby prefectures.[9] Overall, I find that prefectures did react to changes in tariffs on exports in other prefectures in the same province, indicating a competition effect by geographic proximity.

Table C6: Competition between prefectures, geographic proximity

Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ $\sinh^{-1}$(regulation score), 2001–2007		All other prefectures, same prov.			5 closest prefectures		
Δ tariffs on exports, own	-1.60***	-1.35**	-1.45***	-1.35**	-1.32**	-1.54***	-1.32**
2001–2007	(0.50)	(0.50)	(0.51)	(0.51)	(0.55)	(0.51)	(0.55)
Δ tariffs on exports, other pref.		-2.04**		-1.54**			
2001–2007		(0.96)		(0.88)			
Δ $\sinh^{-1}$(regulation score), other pref.			0.36*	0.21			
2001–2007			(0.18)	(0.20)			
Δ tariffs on exports, nearby pref.					-1.13		-1.05
2001–2007					(0.69)		(0.72)
Δ $\sinh^{-1}$(regulation score), nearby pref.						0.12	0.06
2001–2007						(0.13)	(0.13)
Observations	248	245	245	245	248	248	248
R-squared	0.08	0.11	0.11	0.12	0.09	0.08	0.09

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. Column (1) replicates Table 1 Column (5). The mean (s.d.) of Δ $\sinh^{-1}$ (regulation score), 2001–2007 is 1.00 (1.08). The mean (s.d.) of own changes in tariffs on exports, 2001–2007 is –0.18 (0.15). Column (2) controls for the changes in tariffs on exports in all other prefectures in the same province. Column (3) controls for the regulation change in all other prefectures in the same province. Column (4) controls for both. The mean (s.d.) of changes in tariffs on exports in other prefectures in the same province is –0.17 (0.08), and the mean (s.d.) of Δ $\sinh^{-1}$(regulation score) in other prefectures in the same provine is 0.94 (0.43). Columns (5)–(7) repeat the exercise by controlling for the variables in the 5 closest prefectures. The mean (s.d.) of changes in tariffs on exports in the 5 closest prefectures is –0.18 (0.11), and the mean (s.d.) of Δ $\sinh^{-1}$(regulation score) in the 5 closest prefectures is 0.98 (0.57). All columns control for changes in tariffs on imports and inputs and the level of the dependent variable in 2001. The number of observations are smaller in Columns (2)–(4) because Beijing, Shanghai, and Chongqing are both prefectures and provinces.

In addition to focusing on nearby prefectures, there are three other ways to measure a prefecture's exposure to competition with all other prefectures, in terms of changes in tariffs on exports and regulation changes. First, the distance between prefectures can arise from similarities in the industrial composition. The distance between prefecture o and prefecture d is the sum of squared differences in employment shares in each industry:

This is because the metal industry is very high in state-ownership, and as discussed in Section 6.3, state-owned enterprises tend to hire fewer migrants than private firms. The heterogeneous effect is also robust to controlling for individual industry employment shares.

[9] The five nearby prefectures are the five closest prefectures by euclidean distance, calculated from the longitude and the latitude.

$$D_{o,d}^{ind} = \sum_j (EmpShare_{o,j}^{2001} - EmpShare_{d,j}^{2001})^2,$$

where $EmpShare_{i,j}^{2001}$ is the employment share in industry j in prefecture i in 2001, $i \in \{o, d\}$.

Second, the distance can be due to similarities in the population size. The distance between prefecture o and prefecture d is the squared differences in log population in 2001:

$$D_{o,d}^{pop} = (\log(population_o^{2001}) - \log(population_d^{2001}))^2.$$

Third, the distance can come from similarities in per capita GDP:

$$D_{o,d}^{GDP} = (\log(\text{GDP p.c.}_o^{2001}) - \log(\text{GDP p.c.}_d^{2001}))^2.$$

I then construct the weight assigned to each destination prefecture d with respect to an origin prefecture o by taking the inverse of the distance measure as above, combined with the inverse of geographic distance:

$$w_{o,d}^S = \frac{1}{D_{o,d}^S} \cdot \frac{1}{D_{o,d}^{geodist}},$$

where $S \in \{ind, pop, GDP\}$, and $D_{o,d}^{geodist}$ is the travel time between prefecture o and prefecture d in 2000. $T_{ij,2000}$ is from Yang [2017], using the highway and non-highway network in China, with the assumption that the speed of travel is 90 kilometers per hour on highways, 25 kilometers per hour on national and provincial non-highways, and 15 kilometers per hour on local roads.

The change in tariffs on exports in prefectures that compete with prefecture o is measured as

$$TS_o^S = \sum_d \frac{w_{o,d}^S}{\sum_{d'} w_{o,d'}^S} TS_d,$$

and regulation change in the competing prefectures is measured as

$$R_o^S = \sum_d \frac{w_{o,d}^S}{\sum_{d'} w_{o,d'}^S} R_d,$$

where $S \in \{ind, pop, GDP\}$.

I test whether the changes in tariffs on exports and regulation changes in competing prefectures increase a prefecture's incentive to change its own regulation. Table C7 Column (1) includes a prefecture's own changes in tariffs on exports and adds changes in regulation scores in competing prefectures in terms of industrial composition. The coefficient on other prefectures' regulation change is positive but insignificant. Columns (2)–(4) focus on competition by population size. Column (2) includes changes in tariffs on exports of competing prefectures, Column (3) includes regulation changes, and Column (4) includes both. The coefficients are either insignificant or marginally significant. I do the same exercise in Columns (5)–(7), focusing on competition by per capita GDP. I find negative and significant effects of changes in tariffs on exports, and positive and significant of regulation changes. A one-unit change in the tariffs on exports in competing prefectures has almost the same effect as a one-unit change in a prefecture's own tariffs on exports (–1.14 and –1.42 compared to –1.22 and –1.50). The elasticity between a prefecture's own regulation change and the competing prefectures' regulation change is 0.27–0.30.

Table C7: Competition between prefectures, by industrial composition, population size, and income similarity

Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ $\sinh^{-1}$(regulation score), 2001–2007	By ind.	By size of population			By size of GDP p.c.		
Δ tariffs on exports, own	-1.42***	-1.67***	-1.62***	-1.71***	-1.50***	-1.26***	-1.22**
2001–2007	(0.46)	(0.48)	(0.51)	(0.49)	(0.48)	(0.45)	(0.45)
Δ $\sinh^{-1}$(regulation score), other, ind.	0.35						
2001–2007	(0.30)						
Δ tariffs on exports, other, pop.		0.63		0.67*			
2001–2007		(0.38)		(0.38)			
Δ $\sinh^{-1}$(regulation score), other, pop.			0.07	0.08			
2001–2007			(0.09)	(0.09)			
Δ tariffs on exports, other, GDP					-1.42***		-1.14**
2001–2007					(0.44)		(0.43)
Δ $\sinh^{-1}$(regulation score), other, GDP						0.30***	0.27***
2001–2007						(0.09)	(0.09)
Observations	247	247	247	247	247	247	247
R-squared	0.08	0.08	0.08	0.09	0.10	0.12	0.13

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. Column (1) controls for regulation changes in other prefectures, using the distance in hours of travel and closeness of the industrial composition as weights. The mean (s.d.) of Δ $\sinh^{-1}$(regulation score), 2001–2007 is 1.00 (1.08). The mean (s.d.) of own changes in tariffs on exports, 2001–2007 is –0.18 (0.15). The mean (s.d.) of Δ $\sinh^{-1}$(regulation score), in the prefectures with similar industrial composition is 1.11 (0.29). Column (2) controls for changes in tariffs on exports in other prefectures, using the distance in hours of travel and closeness of the population size as weights. Column (3) controls for regulation changes in other prefectures, using the same weights as in Column (2); Column (4) controls for both changes in tariffs on exports and regulation changes in other prefectures. The mean (s.d.) of Δ $\sinh^{-1}$(regulation score), in the prefectures with similar population is 0.99 (0.74) and the mean (s.d.) of changes in tariffs on exports, 2001–2007 is –0.19 (0.15). Columns (5)–(7) repeats the exercise in Columns (2)–(4) using the distance in hours of travel and closeness of GDP p.c. as weights. The mean (s.d.) of Δ $\sinh^{-1}$(regulation score), in the prefectures with similar GDP is 0.95 (0.75) and the mean (s.d.) of changes in tariffs on exports, 2001–2007 is –0.19 (0.11). All columns control for changes in tariffs on imports and inputs and the level of the dependent variable in 2001. The sample restriction is the same as in Table 1 Column (5); in addition, Zhoushan Prefecture is not included in the regression because it is an island and has no distance measure by highway.

Overall, I find that prefectures are competing in regulations with other prefectures that are similar in terms of income. This suggests that prefectures with similar income compete for the same pool of migrants, and there is a significant spillover effect in both changes in tariffs on exports and regulation changes.

C.11 Effects of Regulation Changes on Migrant Flows and Economic Outcomes

C.11.1 Econometric Framework and Identification

I estimate the effect of changes in migration regulation on migrant flows by using the following regression equation:

$$\Delta Y_{it} = \pi_0 + \pi_1 \Delta \sinh^{-1}(\text{regulation score})_{it} + X_{it}\Phi + \zeta_{it},$$

where ΔY_{it} can be the 2000–2010 change of the migrant share of population in prefecture i or the change in the log migrant stock, and π_1 represents how regulation changes affect the outcome variables.

There are several challenges in identifying π_1. First, it is not clear whether the causal relationship goes from migration regulations to migrant flows or the other way around. On the one hand, migration regulation changes can affect migrant amenities, making the prefecture more or less attractive to migrants and leading to bigger or smaller migrant inflows. On the other hand, larger migrant inflows could put pressure on prefecture infrastructure and local employment, leading to regulation changes.

Second, there could be omitted variables that are correlated with both the regulation changes and migrant flows. Suppose that some prefectures have larger growth in pro-migrant sentiment, then it could be the case that both communities and the local government become more migrant-friendly. In this case, the effect of the change in migration regulations will also capture the community sentiment effect.

Third, there could be specific omitted variables related to the fact that changes in tariffs on exports affect both migration regulations and migrant flows. Changes in tariffs on exports affect migrant flows through two channels: directly, through prices (and thus, wages), and indirectly, through migration policies. One must control for changes in tariffs on exports in the regression to identify the regulation effect, but this may not be sufficient. Suppose that regions with larger changes in tariffs on exports also enact more favorable land policies to attract firms, and these firms are able to convince the local government to relax migration restrictions. At the same time, favorable land policies make the region more attractive to workers as well. In this case, without explicitly measuring and controlling for such industrial policies, the estimate of π_1 will be biased.

The last concern relates to the measurement of regulations. The migration regulations I collect may not be the complete set of regulations affecting migrant workers. It is possible that a government enacts a regulation that is not specifically targeted at migrant workers, but at all low-skilled workers in a certain industry. My dataset does not capture such regulations if migrant-related keywords do not show up in the regulation title. The second issue is the coding of migrant-friendliness. I code the migrant-friendliness on a five-point scale, but the actual strength of the regulation could be continuous. In addition, enacting a regulation may not be equivalent to enforcing a regulation. I do not have a prior regarding whether the prefectures with bigger changes in regulation scores enforced the regulations more strictly than prefectures with smaller changes. Overall, if the measurement error is random, the coefficient estimate for π_1 is biased towards zero.

Keeping all of these challenges in mind, I pursue several approaches to estimating the extent to which changes in migrant regulations facilitated increases in migration to regions facing more favorable trade shocks.

C.11.2 Did Migrant Flows Drive Regulation Changes, or Was It the Other Way Around?

The first exercise provides suggestive evidence against reverse causality. I show that changes in migration legislation preceded changes in migrant flows by looking at the timing of the regulation change and the migration flow changes, as well as at the leads and lags. In Table C8, I check the effect of regulation changes in different time periods on migrant flows from 2005 to 2010. Column (1) shows that a one-unit increase in the inverse-hyperbolic-sine-transformed regulation score from

1995 to 2000 (two lagged periods) is related to a 2.28-percentage-point increase in the migrant share of population from 2005 to 2010. In Column (2), I use regulation changes from 2000 to 2005 (one lagged period), and the coefficient on the change in the inverse-hyperbolic-sine-transformed regulation score declines to 1.47. Column (3) uses the contemporaneous regulation change from 2005 to 2010, and the coefficient declines to 0.64. This could be the mechanical effect from the fact that the mean change in the inverse-hyperbolic-sine-transformed regulation scores increases from 0.003 in Column (1) to 1.24 in Column (3). However, when we go to Column (4), although there is still a sizable change in the inverse-hyperbolic-sine-transformed regulation score of 0.6 from 2010 to 2015, there is no longer a positive effect of regulation changes on migrant flows from 2005 to 2010.

Table C8: Regulation changes and migrant flows, lagged, current, and lead

	(1)	(2)	(3)	(4)
Dependent variable:	Δ migrant share of population, 2005–2010			
Δ $\sinh^{-1}$(regulation score), 1995–2000	2.28*			
	(1.18)			
Δ $\sinh^{-1}$(regulation score), 2000–2005		1.47**		
		(0.53)		
Δ $\sinh^{-1}$(regulation score), 2005–2010			0.64**	
			(0.28)	
Δ $\sinh^{-1}$(regulation score), 2010–2015				-1.17***
				(0.43)
Migrant share of population, 2005	0.09	0.05	0.09	0.09
	(0.08)	(0.09)	(0.09)	(0.08)
Log population, 2005	-0.36	-0.54	-0.28	-0.24
	(0.66)	(0.62)	(0.65)	(0.67)
Observations	248	248	248	248
R-squared	0.06	0.07	0.04	0.05
Mean (s.d) of X	0.003 (0.39)	0.41 (0.80)	1.24 (0.95)	0.65 (0.61)

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. The sample restriction is the same as in Table 1 Column (5). The mean value of the change in the migrant share of population from 2005 to 2010 is 5.48 (5.41).

Overall, I find a positive effect of lagged or current regulation changes on migrant flows, but no effect of lead regulation changes. This finding suggests that regulations were indeed binding, and changes in regulations determined migration, rather than being the result of migrant flows.

C.11.3 Changes in Tariffs on Exports, Regulation Changes, and Economic Outcomes: OLS and IV Results

In the following section, I show that OLS and IV estimates of the migration regulation effect are similar and not statistically different from each other. Both changes in tariffs on exports and regulation changes contributed positively to migrant inflows, and also affected wages, employment and local GDP growth.

OLS Results In this section, I discuss how changes in tariffs on exports affected other economic outcomes such as wages, employment, and GDP growth. In Table C9 Column (1), a one-standard-deviation larger decline in tariffs on exports leads to a 2.4% larger increase in wages. This is 2.9%

of the mean and 17% of one standard deviation for changes in wages. Using the estimate of the trade effects in Table 1 Column (5), the regulation effect is 20% of the total trade effect (i.e., $0.02 * (-1.60)/(-0.16) = 0.2$).

Table C9: Changes in tariffs on exports, changes in regulations, and changes in wages, employment, and per capita GDP (2001–2007)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependent variable:		Δ log wage			Δ log GDP p.c.			Δ log total urban emp.	
Δ tariffs on exports	-0.16**		-0.14**	-0.56***		-0.53***	-0.38*		-0.37*
2001–2007	(0.07)		(0.06)	(0.13)		(0.13)	(0.21)		(0.20)
Δ $\sinh^{-1}$(regulation score)		0.02***	0.02***		0.06***	0.05***		0.07**	0.04*
2001–2007		(0.01)	(0.01)		(0.02)	(0.02)		(0.03)	(0.02)
Observations	248	248	248	248	248	248	248	248	248
R-squared	0.31	0.26	0.33	0.27	0.21	0.31	0.22	0.08	0.23
Mean (s.d.) of depend.		0.82 (0.14)			0.86 (0.28)			0.32 (0.39)	

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. All columns control for changes in tariffs on imports and inputs, the log total population in 2001, and the level of the dependent variable in 2001. The mean (s.d.) Δ $\sinh^{-1}$(regulation score), 2001–2007 is 1.00 (1.08), and the mean (s.d.) change in tariffs on exports is –0.18 (0.15). The sample restriction is the same as in Table 1 Column (5).

The effect of regulation changes on wages can go either way, depending the relative size of the increase in local wages and the decrease in migrant wages. My finding of a positive effect of regulation changes on wages is similar to the finding in Lee et al. [2017], where the authors study the effect of the U.S. repatriation of Mexicans in the 1930s on local employment, and they find that the decrease in the number of Mexican workers was associated with small decreases in native employment and increases in native unemployment. Although my results point to the wage margin rather than the employment margin, the finding suggests that an inflow of migrant workers could be beneficial for local workers overall.

The effect of changes in tariffs on exports and the effects of regulation changes are bigger for per capita GDP and total urban employment than for wages. Prefectures with a one-standard-deviation larger decline in tariffs on exports had a 8.4% larger increase in per capita GDP, and a 5.7% larger increase in employment than the small-decline prefectures. This is 9.7% of the mean for changes in per capita GDP, and 17.8% of the mean for changes in employment. The effect of regulation changes on GDP is 14.3% of the total trade effect (i.e., $0.05 * (-1.6)/(-0.56) = 0.143$), and the effect of regulation changes on employment is 16.8% of the total trade effect (i.e., $0.04 * (-1.6)/(-0.38) = 0.168$).

Overall, the effect of changes in tariffs on exports trade on wages, income, and employment is statistically significant and economically large. The effect on per capita GDP is bigger than the effects on wages and employment, potentially capturing other channels through which changes in tariffs on exports affected the economy (through payment to other factors, for example). The regulation channel is significant for wages, per capita GDP, and total urban employment, and the regulation effect is about 14.3% to 20% of the total trade effect.

IV Results To address the concerns on the identification of the migration regulation effect using the OLS regressions, I instrument the regulation changes using the 2000 natural population growth rate. The natural growth rate of the population (birth rate minus death rate) predicts the future population size of a prefecture. It can be a relevant instrument since a higher natural growth rate means that the prefecture will have a more abundant workforce, and the local government is less likely to relax migration restrictions. At the same time, the natural population growth rate is not likely to be correlated with government industrial policies, which is an important potential omitted variable. I test empirically that conditional other 2000 prefecture characteristics, the 2000 natural population growth rate is not correlated with migrant flow from 2000 to 2010.

One might be concerned that given the one-child policy in China, there is no cross-sectional variation in natural population growth rates. As summarized in Zhang [2017], although the policy was applicable throughout China, the actual implementation of the policy vary a lot over time and across regions.[10] In addition, fertility rates and death rates depend on factors such as the local population age structure, food consumption, habits (such as smoking), and environmental pollution, which are not directly managed by government policies.[11] Empirically, among the 248 Chinese prefectures used in the main analysis, the mean natural population growth rate is 5.2 per thousand, with a standard deviation of 2.8.

Table C10: Natural growth rate as an IV for regulation changes, first-stage and IV results

Depend. variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Δ 2001–2007	$\sinh^{-1}$ (reg.	Migrant share of pop.			Wage		GDP p.c.		Employm.	
	score)	OLS	OLS	IV	OLS	IV	OLS	IV	OLS	IV
Δ tariffs on exports	-1.36**	-6.11*	-6.23*	-5.53	-0.14**	-0.11	-0.53***	-0.48***	-0.37*	-0.36*
2001–2007	(0.53)	(3.22)	(3.22)	(3.60)	(0.06)	(0.08)	(0.13)	(0.18)	(0.20)	(0.21)
Δ $\sinh^{-1}$(regulation score)		0.84***	0.87***	1.57	0.02***	0.06	0.05***	0.15	0.04*	0.05
2001–2007		(0.23)	(0.22)	(2.56)	(0.01)	(0.05)	(0.02)	(0.21)	(0.02)	(0.17)
Natural pop. growth rate	-0.09***	-0.06								
2000	(0.03)	(0.21)								
Observations	248	248	248	248	248	248	248	248	248	248
R-squared	0.12	0.16	0.16	0.15	0.33	0.28	0.31	0.19	0.23	0.23
First-state F stat	-	-	-	11		11		5		7
Hausman test p-value		1.0			1.0		1.0		1.0	
Mean (s.d.) of depend.	1.0 (1.1)	6.9 (5.8)			0.8 (0.1)		0.9 (0.3)		0.3 (0.4)	

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. Column (1) has the same specification as in Table 1 Column (5), and adds the natural growth rate of population in 2000. Column (3) has the same specification as in Table 4 Panel A Column (3), and Column (2) adds the natural population growth rate. Column (4) instruments Δ $\sinh^{-1}$(regulation score) with the natural growth rate of population, and the $\sinh^{-1}$(regulation score) in 2001. Columns (5), (7), and (9) are the same as in Table C9 Columns (3), (6), and (9), and Columns (6), (8), and (10) are the corresponding IV regressions. The mean (s.d.) change in tariffs on exports is –0.18 (0.15).

Table C10 Column (1) regresses the change in the inverse-hyperbolic-sine-transformed regulation scores on changes in tariffs on exports as in Table 1 Column (5), controlling for the 2000

[10]For example, Qian [2009], Liu [2014], and Li and Zhang [2017] use different proxies for regional stringency of the one-child policy to study the impact of the policy on various outcomes.

[11]Environmental regulations can affect pollution levels, but it is unlikely that these policies are correlated with migration regulations.

natural growth rate of population. The coefficient for the natural growth rate is negative and statistically significant, meaning that in prefectures with higher natural growth rates, the increase in migrant regulation score is smaller. Column (2) adds the natural population growth rate in the regression of migrant flows on changes in tariffs on exports and regulation changes, and its coefficient is insignificant. I then repeat the OLS regression in Table 4 and Table C9 regarding migrant flows, wages, per capita GDP, and employment, and I also use the 2000 natural growth rate and the 2001 inverse-hyperbolic-sine-transformed regulation score as instruments for the change in the inverse-hyperbolic-sine-transformed regulation scores from 2001 to 2007. Compared with the OLS estimates, the effect of changes in regulation scores on economic outcomes is bigger in the IV regressions. However, the IV standard errors are much bigger, and the difference between the OLS estimates and the IV estimates are not statistically significant according to the Hausman test.

C.11.4 Decomposition of the Migrant Flow

Table 4 classifies migrant flows into short-, medium-, and long-distance categories. As a robustness check, Table C11 and Table C12 use alternative classifications: (1) the purpose of migration in Columns (1)–(4); (2) the time since migrating in Columns (5)–(6); and (3) years of education in Columns (7)–(8). The specifications here are the same as in Table 4 Panel A Column (5). In Table C11, I investigate the effect of changes in tariffs on exports, and in Table C12, I investigate the effect of regulation changes on migrant flows.

Table C11: The relationship between changes in tariffs on exports (2001–2007) and migrant flow in subcategories (2000–2010)

Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ log # of migrants		Purpose of migration			Time since migrated		Years of education	
Subcategory	Work	Family	Marriage	Other	<= 3	> 3	<12	>= 12
Δ tariffs on exports	-0.42*	-0.54**	-0.29	-0.85***	-0.41*	-0.55**	-0.25	-0.59**
2001–2007	(0.24)	(0.22)	(0.29)	(0.25)	(0.21)	(0.24)	(0.19)	(0.28)
Y, 2001	-0.30***	-0.22***	-0.39***	-0.06	-0.25***	0.11*	0.00	-0.08
	(0.05)	(0.04)	(0.05)	(0.06)	(0.05)	(0.06)	(0.04)	(0.06)
Log population, 2001	0.14*	0.09	0.28***	0.08*	0.16**	-0.09	-0.04	0.03
	(0.08)	(0.07)	(0.08)	(0.04)	(0.06)	(0.06)	(0.04)	(0.07)
Observations	248	248	246	248	248	248	248	248
R-squared	0.35	0.13	0.32	0.14	0.28	0.09	0.06	0.08
Mean (s.d) of depend.	1.5 (0.6)	1.2 (0.5)	0.9 (0.5)	-0.4 (0.5)	1.0 (0.5)	0.5 (0.5)	0.4 (0.3)	0.9 (0.4)

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. All columns control for changes in tariffs on imports and inputs. The mean value (s.d.) of changes in tariffs on exports from 2001 to 2007 is –0.18 (0.15). The sample restriction is the same as in Table 1 Column (5).

I find that changes in tariffs on exports and the relaxation of migration restrictions affected people who migrated for work and for family the most and people who migrated for marriage the least. This is a reasonable result, since the regulations were mostly work-related. Regulation changes had bigger effects in the later period (migrated in the nearest three years from the time of the survey) than the current period (migrated in more than three years ago from the time of the survey). This finding is consistent with Table C8: regulations take time to impact migrant flows.

Finally, both changes in tariffs on exports and the regulation changes affected the migrants with more than 12 years of education the most. In the 2000–2010 period, the medium- and long-distance migrant flows increased a lot, and it seems that more-educated migrants were the driving force.

Table C12: The relationship between regulation changes (2001–2007) and migrant flow in subcategories (2000–2010)

Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ log # of migrants		Purpose of migration			Time since migrated		Year of education	
Subcategory	Work	Family	Marriage	Other	<= 3	> 3	<12	>= 12
Δ $\sinh^{-1}$(regulation score)	0.10***	0.08**	0.04	0.09**	0.09***	0.05*	0.04	0.10***
2001–2007	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Y, 2001	-0.32***	-0.24***	-0.42***	-0.10	-0.29***	0.11**	-0.01	-0.12**
	(0.04)	(0.04)	(0.05)	(0.07)	(0.03)	(0.05)	(0.04)	(0.05)
Log population, 2001	0.17**	0.09	0.27***	0.05	0.18***	-0.07	-0.01	0.06
	(0.08)	(0.06)	(0.07)	(0.04)	(0.06)	(0.05)	(0.04)	(0.07)
Observations	248	248	246	248	248	248	248	247
R-squared	0.35	0.13	0.30	0.03	0.29	0.06	0.01	0.06
Mean (s.d) of depend.	1.5 (0.6)	1.2 (0.5)	0.9 (0.5)	-0.4 (0.5)	1.0 (0.5)	0.5 (0.5)	0.4 (0.3)	0.9 (0.4)

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. The mean value (s.d.) of Δ $\sinh^{-1}$(regulation score) from 2001 to 2007 is 1.0 (1.1). The sample restriction is the same as in Table 1 Column (5).

C.11.5 Emigration Instead of Immigration

Table C13: The effects of changes in tariffs on exports (2001–2007) and regulation changes (2001–2007) on emigration (2000–2010)

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	Δ out-migration share			Δ log number of emigrants		
Δ tariffs on exports, 2001–2007	4.20		3.93	-0.02		0.04
	(3.24)		(3.23)	(0.15)		(0.15)
Δ $\sinh^{-1}$(regulation score), 2001–2007		-0.44	-0.20		0.03	0.04
		(0.46)	(0.42)		(0.02)	(0.02)
Observations	248	248	248	248	248	248
R-squared	0.04	0.01	0.04	0.67	0.67	0.68
Mean (s.d.) of Y	11.3 (6.4)			1.1 (0.5)		

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. Dependent variables are changes from 2000 to 2010. All columns control for changes in tariffs on imports and inputs, the log total population and the level of the dependent variable in 2000. The mean (s.d.) Δ $\sinh^{-1}$(regulation score), 2001–2007 is 1.00 (1.08), and the mean (s.d.) changes in tariffs on exports is –0.18 (0.15). The sample restriction is the same as in Table 1 Column (5).

The 2000 and 2010 censuses also collected information on emigration, since each household was asked to report the number of family members who left their Hukou location for more than six months. Table C13 replicates the results in Table 4 Panel A Columns (1)–(3) by replacing the immigration share of population with emigration share of population (Columns 1–3) or with in log number of out-migrants (Columns 4–6). Overall, there is no consistent significant effect of either changes in tariffs on exports or regulation changes on emigration.

C.11.6 Migrant Supply

Table 4 shows the effects of changes in tariffs on exports and changes in regulations on migrant flow. I further investigate the heterogeneous effect with respect to migrant supply in Table C14. The potential supply of migrants can affect the responsiveness of migrant flow to changes in tariffs on exports and regulation changes. For prefecture o, the distance-weighted agricultural population is

$$\log(agrPOP)_o^{2001} = \sum_d \frac{w_{o,d}}{\sum_{d'} w_{o,d}} \log(agrPOP)_d^{2001},$$

where $w_{o,d} = \frac{1}{D_{geodist_{o,d}}}$, which is inverse of travel time between prefecture o and prefecture d in 2000, and $\log(agrPOP)_d^{2001}$ is the log agricultural population in prefecture d in 2001.

Table C14: Interaction effects of migrant supply and migrant demand

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependent variable:	Δ migrant share of pop			Δ log # of migr. short dist.			Δ log # of migr. medium dist.		
Δ tariffs on exports	-6.53*	-23.81	-6.57*	-1.01***	-4.31	-1.04***	-0.39**	2.24	-0.45**
2001–2007	(3.44)	(30.42)	(3.45)	(0.27)	(4.38)	(0.26)	(0.18)	(6.46)	(0.16)
Δ $\sinh^{-1}$(regulation score)	0.88***	0.87***	-4.07	0.18***	0.17***	-0.49	0.05*	0.06*	-1.48*
2001–2007	(0.22)	(0.23)	(5.25)	(0.04)	(0.04)	(0.51)	(0.03)	(0.03)	(0.85)
Log (agr. pop.), 2001	-0.63	-0.21	-1.05	-0.42**	-0.35	-0.51**	-0.00	-0.03	-0.11
	(0.79)	(1.20)	(0.89)	(0.13)	(0.16)	(0.14)	(0.09)	(0.12)	(0.09)
Δ tariffs on exports		1.35			0.25			-0.16	
×Log (agr. pop.)		(2.49)			(0.41)			(0.39)	
Δ $\sinh^{-1}$(regulation score)			0.38			0.05			0.09*
×Log (agr. pop.)			(0.41)			(0.04)			(0.05)
Observations	247	247	247	238	238	238	248	248	248
R-squared	0.16	0.16	0.16	0.28	0.28	0.28	0.59	0.59	0.60
Mean (s.d.) of depend.	6.9 (5.7)			-0.8 (0.8)			1.6 (0.7)		

Note: Standard errors are clustered at the province level. *** p<0.01, ** p<0.05, * p<0.1. The outcome variables are changes from 2000 to 2010. All columns control for changes in tariffs on imports and inputs, the log total population and the level of the dependent variable in 2000. Columns (1)–(3) use the weighted average agricultural population. Columns (4)–(6) use the agricultural population in the same prefecture. Columns (7)–(9) use the agricultural population in the same province. The mean (s.d.) Δ $\sinh^{-1}$(regulation score), 2001–2007 is 1.0 (1.1), the mean (s.d.) change in tariffs on exports is –0.18 (0.15). The sample restriction is the same as in Table 1 Column (5). Columns (1)–(3) do not include Zhoushan Prefecture due to the missing distance via highway.

Columns (1)–(3) use the change in the migrant share of population as the outcome and control for agricultural population, measured as above. In addition, Column (2) adds the interaction between trade shocks and agricultural population, and Column (3) adds the interaction between the regulation change and agricultural population. I find no significant effect either on the agricultural population or on the interaction.

Columns (4)–(6) investigate the effect on short-distance migrant flows, where migrants move within a prefecture. Thus, I use the agricultural population in the same prefecture. There is no significant interaction effect.

Column (7)–(9) show the effect on medium-distance migrant flows, where migrants move within a province across different prefectures. I use the agricultural population in the whole province as the measure for the potential pool of migrant supply. I find a positive interaction effect between the regulation change and migrant supply: a prefecture that is part of a province with a lot of agricultural population has a bigger inflow of migrant workers once the regulation is relaxed.

References

David H. Autor, David Dorn, and Gordon H. Hanson. The China Syndrome: Local Labor Market Effects of Import Competition in the United States. *American Economic Review*, 103(6):2121–68, 2013. doi: 10.1257/aer.103.6.2121.

Kirill Borusyak, Peter Hull, and Xavier Jaravel. Quasi-experimental shift-share research designs. Working Paper 24997, National Bureau of Economic Research, September 2018.

Miroslav Kubat. *An Introduction to Machine Learning*. Springer, 2017.

Jongkwan Lee, Giovanni Peri, and Vasil Yasenov. The Employment Effects of Mexican Repatriations: Evidence from the 1930's. Working Paper 23885, National Bureau of Economic Research, September 2017.

Bingjing Li and Hongliang Zhang. Does Population Control Lead to Better Child Quality? Evidence from China's One-Child Policy Enforcement. *Journal of Comparative Economics*, 45(2):246–260, 2017.

Haoming Liu. The Quality-Quantity Trade-Off: Evidence from the Relaxation of China's One-Child Policy. *Journal of Population Economics*, 27(2):565–602, 2014.

Nancy Qian. Quantity-Quality and the One Child Policy: The Only-Child Disadvantage in School Enrolment in Rural China. Technical report, National Bureau of Economic Research, 2009.

Yang Yang. Transport Infrastructure, City Productivity Growth and Sectoral Reallocation: Evidence from China. Working paper, January 2017.

Junsen Zhang. The Evolution of China's One-Child Policy and Its Effects on Family Outcomes. *Journal of Economic Perspectives*, 31(1):141–60, 2017.